

#1 · Organization-Wide Planning and Deployment

What is Six Sigma and where did it start?

Answer

Six Sigma is a methodology for reducing variation and defects, with the statistical goal of no more than 3.4 defects per million opportunities. It was developed at Motorola in 1986 by Bill Smith and was later championed at General Electric by Jack Welch. A hospital that cuts medication errors from 5 per 1,000 doses to 3 per million is working at a Six Sigma level.

#2 · Organization-Wide Planning and Deployment

How do lean and Six Sigma work together?

Answer

Lean removes waste and speeds up flow, while Six Sigma reduces variation and defects. They are complementary because lean reveals the process steps that matter and Six Sigma makes those steps consistent. A warehouse can use lean to cut travel distance and Six Sigma to cut the rate of mislabeled shipments.

#3 · Organization-Wide Planning and Deployment

What screening criteria are used to select a Six Sigma project?

Answer

Screening criteria are the filters a leadership team applies before chartering a project, such as link to strategy, size of the financial or customer benefit, data availability, feasibility, resource needs, and risk. The chosen project must matter to the business and be measurable. A tool crib rejects a project that would save 2,000 dollars but consume a Black Belt for a year, because the return does not justify the resources.

#4 · Organization-Wide Planning and Deployment

What is a SWOT analysis and how does it support strategic planning?

Answer

SWOT stands for strengths, weaknesses, opportunities, and threats, with strengths and weaknesses looking inside the organization and opportunities and threats looking outside it. The analysis helps leaders choose which Six Sigma and lean initiatives to fund. A hospital sees a strength in its skilled nurses, a weakness in paper records, an opportunity in telehealth, and a threat from a new competitor clinic.

#5 · Organization-Wide Planning and Deployment

How does contingency planning differ from business continuity planning?

Answer

Contingency planning prepares a specific response for a risk event, such as a supplier failure, while business continuity planning keeps the most critical operations running during a major disruption. Both are part of strategic deployment. A plant keeps a second approved supplier as contingency and keeps a manual order process as continuity for a system outage.

#6 · Organization-Wide Planning and Deployment

What are the roles of a champion, a sponsor, and a process owner?

Answer

A champion is a senior leader who promotes the project and clears organizational barriers, and a sponsor provides the funding and the formal authority for the work. A process owner takes responsibility after the project closes and keeps the new process running. In a bank, the operations director champions a loan cycle time project, and the branch manager becomes the process owner at handoff.

#7 · Organization-Wide Planning and Deployment

How do the belt roles differ in a Six Sigma deployment?

Answer

A Green Belt leads smaller projects part time while keeping a regular job, a Black Belt leads larger projects full time and coaches Green Belts, and a Master Black Belt is a full time expert who trains and mentors Black Belts and works on deployment strategy. The belts form a support chain rather than a rank ladder. One Master Black Belt might train ten Black Belts, each of whom coaches five Green Belts.

#8 · Organization-Wide Planning and Deployment

What change management techniques help a Six Sigma project succeed?

Answer

Change management uses stakeholder analysis to find who is affected and how much influence they have, a readiness assessment to judge how prepared the organization is, and a communication plan to explain the purpose and the progress. Common causes of failure include weak management support and missing resources. A plan that explains the reason for a new standard work step before training sees far less resistance than one that simply posts the new instruction.

#9 · Organization-Wide Planning and Deployment

What is the difference between a business case and a project charter?

Answer

The business case argues why the project is worth doing, including the cost, the benefit, and the risk of doing nothing. The charter then authorizes the work by naming the problem, the scope, the goal, the team, and the timeline.

#10 · Organization-Wide Planning and Deployment

Why does a deployment need one standard format for reporting project benefits?

Answer

A standard format makes every project state its baseline, its target, its measure, its verified saving, and its control plan in the same way, so leaders can compare and audit them. Without it, one team reports optimistic estimates and another reports hard savings, and the portfolio total becomes meaningless.

#11 · Organization-Wide Planning and Deployment

Why should Six Sigma projects be planned inside the annual business planning cycle?

Answer

Tying project selection to the business plan connects the improvement portfolio to the goals leaders have already committed to fund. A company that starts projects from whoever complains loudest ends up with a full roster and no measurable effect on its strategy.

#12 · Organization-Wide Planning and Deployment

What does a steering committee do that a project team does not?

Answer

A steering committee sets priorities, approves resources, resolves cross-functional conflicts, and reviews the portfolio, while the project team does the analysis and the improvement work. The committee meets on a slower cycle, so the team is not waiting on it for routine decisions.

#13 · Organization-Wide Planning and Deployment

How does a deployment communicate its purpose across the organization?

Answer

Leaders repeat a short, consistent message about why the organization is deploying Six Sigma, what it expects to change, and how progress will be reported. A plant that hears the same story from the plant manager, the champion, and the finance lead is far less likely to treat the work as the flavor of the month.

#14 · Organization-Wide Planning and Deployment

What separates an enterprisewide deployment from a cluster of isolated projects?

Answer

An enterprisewide deployment has a shared method, trained belts at several levels, a project selection process, and a review that tracks benefits against strategy. Isolated projects can still save money, but they fade when the champion moves on because no system carries the method forward.

#15 · Organization-Wide Planning and Deployment

How does a team choose between a rapid improvement event and a full DMAIC project?

Answer

A rapid event fits a contained problem with a known cause and a fix that can be tested in days, while a DMAIC project fits a chronic problem whose cause is unknown and whose data work will take weeks. A team facing a messy handoff with no clear cause should not book a one-week event and hope.

#16 · Organization-Wide Planning and Deployment

What should a Black Belt do when a project loses its executive sponsor midway?

Answer

The Belt should report the risk, restate the business case, and ask the steering group to assign a new sponsor with the authority to release resources. Continuing quietly without a sponsor usually means the team cannot get decisions or protected time, and the project stalls at the first conflict.

#17 · Organization-Wide Planning and Deployment

What is the 1.5 sigma shift convention?

Answer

The 1.5 sigma shift is the convention that a process drifts about 1.5 standard deviations over the long run, so short-term capability is reported at a higher sigma level than long-term performance. Under the shift, a process with six sigma short-term capability is expected to produce about 3.4 defects per million opportunities over the long term. Sigma tables apply the shift, which is why 3.4 defects per million is called six sigma even though the long-term Z value behind it is 4.5.

#18 · Organization-Wide Planning and Deployment

What is the main goal of lean?

Answer

Lean removes waste and improves flow so that work moves through a process with the shortest possible delay. Value is defined by the customer, and any step the customer would not pay for is a candidate for removal. A hospital that stops moving patients between three waiting rooms is applying lean thinking.

#19 · Organization-Wide Planning and Deployment

How does organizational structure affect a Six Sigma deployment?

Answer

Structure is the shape of the organization, so a functional structure that rewards each department for its own numbers makes cross-department improvement harder than a process structure does. A department measured only on its own cost has little reason to help the next department, even when the whole process would improve. The countermeasures are process owners with real authority, cross-functional teams with named members, and measures that describe the whole process instead of one department.

#20 · Organization-Wide Planning and Deployment

What is a lean Six Sigma maturity model?

Answer

A maturity model describes how deeply improvement thinking is built into an organization, from scattered one-off projects to a fully embedded management system. It helps leaders see which capability to build next, such as project selection, coaching, or standard work. A company at a low level may run a few Black Belt projects, while a mature one ties every project to strategy and tracks benefits every month.

#21 · Organization-Wide Planning and Deployment

How does organizational culture affect a Six Sigma deployment?

Answer

Culture is how people behave when nobody is watching, so a blaming culture drives people to hide defects and report numbers late, which makes the data unreliable. A learning culture treats a defect as information, and leaders ask what the process allowed to happen rather than who to blame. The countermeasures are visible leadership behavior, measures that describe the process rather than one person, and protection for the people who surface bad news.

#22 · Organization-Wide Planning and Deployment

What is stakeholder analysis, and how is it used on a Six Sigma project?

Answer

Stakeholder analysis lists every person or group who can affect the project or will be affected by it, then records each one's interest, influence, present stance, and what that stakeholder needs in order to move. The team uses the list to decide who must be consulted, who needs early communication, and which objections must be answered before implementation. A scheduling project at a plant touches operators, supervisors, maintenance, planning, sales, and possibly a union, and each group needs a message from someone it trusts.

#23 · Organization-Wide Planning and Deployment

What does executive leadership do in a Six Sigma deployment?

Answer

Executive leadership sets direction by treating improvement as a business strategy, chooses the goals the projects must serve, commits people and money, and removes barriers that only executives can remove. Leaders also review the portfolio on a fixed cadence and hold managers accountable for results. Finance supports the work by agreeing the baseline and the counting rules before the project starts and confirming the result at the close.

#24 · Organization-Wide Planning and Deployment

What are common causes of Six Sigma failure, and how does change management help?

Answer

Common failure causes include weak management support, missing resources, poor project selection, unclear goals, and a culture that resists change. Change management uses stakeholder analysis, readiness assessment, communication plans, and proactive support to lower that resistance. A project that saves money but ignores the people who must run the new process often fails after the team leaves.

#25 · Organization-Wide Planning and Deployment

What do recovery time objective and recovery point objective mean in business continuity planning?

Answer

The recovery time objective is the longest the process may stay down, and the recovery point objective is how much data the business can afford to lose, measured as the time between the last backup and the disruption. A payment system might set a recovery time objective of 4 hours and a recovery point objective of 5 minutes.

#26 · Organization-Wide Planning and Deployment

How can speeding up one process overload another in the same system?

Answer

Processes are linked, so a faster upstream step can push work into a downstream step that cannot absorb it, which moves the bottleneck instead of removing it. A shorter credit check can flood the underwriting queue, so the team checks the whole chain before declaring a win.

#27 · Organization-Wide Planning and Deployment

What does an organizational readiness assessment check before a process change?

Answer

A readiness assessment checks whether people understand why the change is coming, have the skills and tools to work the new way, see a benefit, and hold the authority to act. A hospital that has the new software but no staff training is not ready, even when the technology is installed.

#28 · Organization-Wide Planning and Deployment

What is leader standard work in a lean deployment?

Answer

Leader standard work is a written routine of daily and weekly checks that leaders perform to support the new way of working, such as a gemba walk, a review of yesterday's problems, and a check of the visual board. A plant manager who walks the line at the same time each day and asks the same three questions keeps the improvement visible after the project team leaves.

#29 · Organization-Wide Planning and Deployment

What is policy deployment, and how does it connect goals to projects?

Answer

Policy deployment, also called hoshin kanri, is a planning process that moves a few top goals down through each level of the organization and back up again, so every project has a clear owner and measure. A company goal of cutting claims cost by 10 percent becomes a plant target, then a project charter, then a daily metric.

#30 · Organization-Wide Planning and Deployment

Why should a Six Sigma portfolio mix quick wins with longer projects?

Answer

A mix gives the organization early visible results while the harder projects run, which keeps support alive through the long DMAIC cycle. A portfolio might pair two 8-week projects that cut obvious rework with one 9-month project that redesigns a core process.

#31 · Organization-Wide Planning and Deployment

Why does a Six Sigma deployment limit how many projects run at once?

Answer

Each active project needs a belt, a team, data, and sponsor attention, so too many open projects spread those scarce resources and all of them slow down. A plant with two Black Belts may cap the active list at four projects and hold the rest in a queue.

#32 · Organization-Wide Planning and Deployment

How does an organization choose the first site for a new improvement approach?

Answer

The first site should have a willing leader, a visible process, and enough data to show a result, because the pilot becomes the proof the rest of the organization sees. A company rolling out a new method may start at one plant with a supportive manager rather than the plant with the worst numbers.

#33 · Organization-Wide Planning and Deployment

What are the five phases of DMAIC, and what does each phase deliver?

Answer

DMAIC stands for Define, Measure, Analyze, Improve, and Control. Define produces the charter and the voice of the customer, Measure produces the baseline data and the measurement system check, Analyze produces a verified root cause, Improve produces a tested solution, and Control produces the control plan and the handoff to the process owner. A billing team might define a 9 percent error rate, measure where it starts, analyze the address entry step as the source, improve it with a lookup rule, and control it with a weekly chart.

#34 · Organization-Wide Planning and Deployment

What is cost of poor quality, and how does it help rank candidate projects?

Answer

Cost of poor quality adds up the money a process wastes through scrap, rework, warranty claims, and lost goodwill, which are internal and external failure costs. A plant with 400,000 dollars of scrap, 150,000 dollars of rework, and 250,000 dollars of warranty cost carries 800,000 dollars of poor quality cost, which is 8 percent of a 10 million dollar revenue base. Appraisal costs such as inspection are tracked separately, and projects that attack the largest failure buckets usually earn funding first.

#35 · Organization-Wide Planning and Deployment

What does a deployment plan for a Six Sigma initiative contain?

Answer

A deployment plan lists the projects in the pipeline, the belts assigned to each one, the training waves, the budget, the review cadence, and the measures that show whether the initiative is working. A manufacturer might plan four Black Belt projects and twelve Green Belt projects for the year, with monthly reviews by the steering team. The plan ties each project to a business goal so the initiative is not a set of unrelated studies.

#36 · Organization-Wide Planning and Deployment

Why must leadership coach belts and managers at several levels, and what does that coaching cover?

Answer

A belt learns the tools in class but needs help applying them to a real process with real politics. Coaching covers project scoping, data access, stakeholder resistance, and how to read a control chart with the process owner. When a sponsor coaches a new Black Belt through the first toll gate review, the belt learns how the organization judges evidence rather than only how to run the arithmetic.

#37 · Organization-Wide Planning and Deployment

How are intangible benefits handled in a Six Sigma business case?

Answer

Intangible benefits are real gains that are hard to convert into cash, such as higher customer satisfaction, lower employee turnover, or reduced safety exposure. A project that cuts a complaint rate from 6 percent to 2 percent may also lift repeat purchases, and the team describes that effect without claiming a precise dollar figure. The business case keeps hard savings and intangibles in separate lines so reviewers can see which claims are auditable.

#38 · Organization-Wide Planning and Deployment

How does a deployment communication plan differ from a team communication plan?

Answer

A deployment plan speaks to the whole organization about why the initiative matters, what will change, and where people can raise concerns. It segments audiences such as executives, middle managers, process owners, and front line staff, and it repeats the message on a schedule. A team communication plan is narrower and covers only the project team meetings, decisions, and status updates.

#39 · Organization-Wide Planning and Deployment

What does a Master Black Belt do that a Black Belt does not?

Answer

A Master Black Belt trains and mentors belts, reviews project charters across the deployment, guards the technical quality of the tools, and helps leadership select projects. A Black Belt leads one project at a time and does the analysis, while a Master Black Belt may review twelve projects a month and step in when a belt is using the wrong test. The Master Black Belt is the technical conscience of the deployment.

#40 · Organization-Wide Planning and Deployment

What is a project pipeline, and who keeps it full?

Answer

A project pipeline is the ranked list of candidate improvement projects that a deployment draws from, sometimes called the project hopper. Champions and process owners feed it with problems surfaced by scorecards, customer complaints, and audits, and the steering team ranks the candidates and assigns belts. A healthy pipeline holds more candidates than there are belts, so a project that stalls can be replaced without idling the belt.

#41 · Organizational Process Management and Measures

How can a Six Sigma project affect customers, suppliers, and other stakeholders?

Answer

A project can change what customers receive, what suppliers must deliver, and how employees work, so each group should be considered before changes start. Some effects are positive, such as shorter lead times, and some are negative, such as tighter incoming inspection for a supplier. A retailer that reduces order errors improves the customer experience but may add a new labeling requirement for its suppliers.

#42 · Organizational Process Management and Measures

What are the main types of benchmarking?

Answer

Competitive benchmarking compares an organization with direct competitors, collaborative benchmarking shares practices with noncompeting partners, best practice benchmarking studies any organization that performs a process well, and breakthrough benchmarking looks for radical new approaches. The type chosen decides who supplies the comparison data. An airline might benchmark its boarding process against a hotel check-in process, which is a best practice study.

#43 · Organizational Process Management and Measures

What is a balanced scorecard?

Answer

A balanced scorecard tracks performance across four perspectives: financial, customer, internal process, and learning and growth. Each measure can be traced upward to an organizational strategy, which is called line of sight. A distribution center may pair its cost per order with an on-time delivery rate, a safety measure, and an employee training completion rate so one view does not hide a problem.

#44 · Organizational Process Management and Measures

What is the difference between a key performance indicator and a key behavior indicator?

Answer

A key performance indicator (KPI) measures an outcome the business cares about, such as on-time delivery or cost per unit, while a key behavior indicator (KBI) measures an action that drives the outcome, such as the share of calls that follow the standard script. KPIs report what already happened and KBIs point at what is happening now. A call center watches both its first-call resolution rate and the rate at which agents complete the diagnostic checklist.

#45 · Organizational Process Management and Measures

What is an objectives and key results (OKR) framework?

Answer

An OKR pairs a short qualitative objective with a small set of measurable key results that show whether the objective is being met. The objective states the direction and the key results state the finish line. A quality team might set the objective of cutting customer escalations with key results of a 40 percent drop in escalations and a 30 percent drop in repeat contacts by the end of the quarter.

#46 · Organizational Process Management and Measures

How do leading indicators differ from lagging indicators?

Answer

A leading indicator changes before the result and helps predict it, while a lagging indicator reports what already happened. Teams use leading indicators to act early instead of explaining a miss afterward. Preventive maintenance completed on schedule is a leading indicator, and unplanned downtime hours are a lagging indicator, so a plant tracks both.

#47 · Organizational Process Management and Measures

What is net present value and how is it used to judge a project?

Answer

Net present value discounts all future cash flows from a project back to today and subtracts the initial investment, so a positive net present value means the project adds value at the chosen discount rate. A project that costs 100,000 dollars and returns 45,000 dollars a year for three years at a 10 percent discount rate has a net present value of about 11,908 dollars. The three discounted returns are 40,909, 37,190, and 33,809 dollars.

#48 · Organizational Process Management and Measures

How do ROI, cost benefit analysis, and hard versus soft benefits differ?

Answer

Return on investment divides the net gain by the cost, cost benefit analysis compares all the costs against all the benefits of an option, and the split between hard and soft benefits separates real profit and loss savings from cost avoidance. Hard savings such as reduced overtime appear on the income statement, while soft savings such as avoided rework are estimates. A project that spends 50,000 dollars to save 80,000 dollars has a return on investment of 60 percent.

#49 · Organizational Process Management and Measures

What is the difference between a balanced scorecard and a dashboard of KPIs?

Answer

A balanced scorecard is a small, linked set of measures chosen to balance financial, customer, process, and learning views against strategy. A dashboard can show twenty operational numbers without any statement of how they connect, which makes it a monitoring tool rather than a strategy tool.

#50 · Organizational Process Management and Measures

How does a team decide which measures belong on a project scorecard?

Answer

A scorecard holds the primary measure that defines success, the secondary measures that guard against trade-offs, and a balance measure that catches harm elsewhere. A project to cut cost might track cost per unit as primary, on-time delivery as secondary, and customer complaints as the balance measure.

#51 · Organizational Process Management and Measures

How does a team set a baseline before an improvement project?

Answer

A baseline is the measured performance of the process before the change, collected over enough time to cover normal variation. Without a baseline the team cannot prove improvement, because a project that starts measuring only after the fix cannot separate the gain from the season.

#52 · Organizational Process Management and Measures

What is an output measure, and how does it differ from a process measure?

Answer

An output measure describes what the process delivers to the customer, such as on-time delivery or defect rate. A process measure describes the work inside the process, such as the share of orders entered with a complete address, and it usually moves first because it is closer to the daily work.

#53 · Organizational Process Management and Measures

What is shareholder value, and how can a quality gain support it?

Answer

Shareholder value is the worth of the owners' stake, driven by profit, growth, and the risk investors attach to future cash flows. A quality gain supports it by lowering waste, protecting revenue through fewer defects, and reducing the chance of recalls or lost customers.

#54 · Organizational Process Management and Measures

How can a project change the relationship with a key supplier?

Answer

A project can change what the supplier must deliver and how it is measured, so the team should bring the supplier in before the new requirement is frozen. A defect project that tightens an incoming inspection spec without telling the supplier usually creates rejections and delays instead of a fix.

#55 · Organizational Process Management and Measures

How is a KPI cascaded from a corporate goal down to a work team?

Answer

A cascade takes one corporate goal, breaks it into the results each level must deliver, and assigns a measure to each level so the numbers add up. If corporate wants 98 percent on-time delivery, the plant might own schedule adherence and planning might own forecast accuracy, because both drive the same result.

#56 · Organizational Process Management and Measures

What is the difference between customer satisfaction and customer loyalty?

Answer

Satisfaction records how a customer judges a recent experience, while loyalty records whether the customer comes back and recommends the organization. A customer can be satisfied with a repair and still switch next time for price, which is why loyalty measures predict revenue better than a single satisfaction score.

#57 · Organizational Process Management and Measures

What is a net promoter score, and how is it calculated?

Answer

A net promoter score asks customers how likely they are to recommend the company on a 0 to 10 scale, then subtracts the percentage of detractors from the percentage of promoters. Scores of 9 or 10 count as promoters, scores of 7 or 8 count as passives, and scores of 0 to 6 count as detractors. If 100 customers answer with 60 promoters and 15 detractors, the score is 60 minus 15, or 45.

#58 · Organizational Process Management and Measures

What is a strategy map, and how does it link the four scorecard perspectives?

Answer

A strategy map draws the cause and effect chain that connects the four balanced scorecard perspectives, from learning and growth to internal process to customer results to financial results. It turns a list of measures into a testable story, so leaders can ask whether a training measure is really expected to lift customer satisfaction. A hotel chain might link staff training to faster check-in, then to higher guest satisfaction, then to repeat bookings and revenue.

#59 · Organizational Process Management and Measures

What is the difference between a KPI and a KBI?

Answer

A key performance indicator measures an outcome the organization cares about, such as defect rate or on-time delivery. A key behavior indicator measures the actions that drive that outcome, such as how often leaders complete a standard work check. Tracking both shows whether people are doing the behaviors that should move the result.

#60 · Organizational Process Management and Measures

What is an OKR?

Answer

An objective and key results pair sets one clear, ambitious objective and a small set of measurable key results that show progress. The objective states the direction, and the key results are numbers with a deadline. A quality team might set an objective to cut customer complaints and track key results such as a 30 percent drop in complaints this quarter.

#61 · Organizational Process Management and Measures

What is the difference between leading and lagging indicators?

Answer

Leading indicators predict future performance, while lagging indicators confirm what already happened. Training completion and first-pass yield can lead, while customer complaints and warranty cost lag. A team that watches only lagging measures learns about problems too late to prevent them.

#62 · Organizational Process Management and Measures

What does line of sight mean for performance measures?

Answer

Line of sight means every measure at the team level connects upward to a larger organizational goal. A call center measure such as first-call resolution should link to customer loyalty and then to revenue growth. Without that link, teams can improve a local number that does not help the business.

#63 · Organizational Process Management and Measures

What financial measures do Black Belts use to justify a project?

Answer

Net present value discounts future cash flows to today so projects of different sizes can be compared, return on investment divides net benefit by cost, and cost benefit analysis compares total benefits with total costs. Hard costs come straight from profit and loss statements, while soft costs are estimates such as avoided cost or freed capacity. A project that costs 100,000 dollars and returns 120,000 dollars has a simple return on investment of 20 percent.

#64 · Organizational Process Management and Measures

How can a Six Sigma project affect stakeholders?

Answer

A project can change what customers receive, what suppliers must deliver, and how employees do their work, so each group can gain or lose. Customers may see better quality or lower cost, suppliers may face new specifications, and workers may face new roles. A Black Belt maps these effects early so that no group is surprised after launch.

#65 · Organizational Process Management and Measures

What does a benefit cost ratio of 1.4 mean in a project review?

Answer

The benefit cost ratio divides the present value of the benefits by the present value of the costs, so 1.4 means \$1.40 of benefit for every \$1.00 of cost. A project with \$140,000 in discounted benefits and \$100,000 in discounted costs has a ratio of 1.4, and any ratio above 1 clears the first screen.

#66 · Organizational Process Management and Measures

How does a team confirm that a leading indicator is worth managing?

Answer

The indicator must move before the lagging result and show a stable relationship with it over time, so the team checks the link with historical data before it manages to the number. A plant tracks daily overdue orders only after showing that the count predicts the monthly on-time score.

#67 · Organizational Process Management and Measures

How does an organization measure supplier performance?

Answer

A supplier scorecard rates each supplier on quality, delivery, cost, and responsiveness, using agreed definitions and a regular review rhythm. A plant that tracks a supplier's on-time rate and defect rate together can show why the low price supplier is not the cheapest overall.

#68 · Organizational Process Management and Measures

Why can cutting inventory improve cash flow without raising profit?

Answer

Inventory is cash tied up in goods, so reducing it releases cash even when the income statement does not change, and the release is a one-time benefit rather than an ongoing saving. A plant that drops inventory from \$2 million to \$1.5 million frees \$500,000 in cash but does not add \$500,000 to profit.

#69 · Organizational Process Management and Measures

How does productivity differ from efficiency?

Answer

Efficiency compares the output actually produced with the output the process could produce from the same input, while productivity compares output with a unit of input such as labor hours or machine hours. A line can run at its standard rate and still have low productivity if it needs far more labor hours per unit than a newer line. Building a product nobody wants is a problem of effectiveness rather than productivity, because the physical output per unit of input can still be high.

#70 · Organizational Process Management and Measures

What can an on-time delivery measure hide?

Answer

A simple on-time percentage can hide partial shipments, expedited freight, and orders that were rescheduled to look on time, so the measure flatters the process. A team watching the promise date rather than the original request date can report 98 percent on time while customers still wait longer than they were told.

#71 · Organizational Process Management and Measures

Why do two departments need one written definition for a shared measure?

Answer

A shared measure only works when both groups count the same events the same way, or each side reports a different number and the argument replaces the improvement work. Sales and operations must agree on when an order counts as late before either group can reduce lateness.

#72 · Organizational Process Management and Measures

How does a yield loss change the profit per good unit?

Answer

A yield loss spreads the cost of the scrapped units over the good units that remain, so the effective cost per good unit rises. If 100 units cost \$6 each to make and only 90 are good, the cost per good unit is \$600 divided by 90, which is \$6.67.

#73 · Organizational Process Management and Measures

What are the steps of a benchmarking study?

Answer

The team plans what to measure and whom to compare against, collects data in the same way from every source, analyzes the gap, and adapts the findings to its own process. A hospital might benchmark its medication reconciliation time against three peer hospitals and learn that two of them use a single shared checklist. The last step matters most, because a number copied without adapting the practice behind it changes nothing.

#74 · Organizational Process Management and Measures

How is a benchmark gap turned into an improvement target?

Answer

The team subtracts its current performance from the benchmark to size the gap, then sets a target it can defend with the time and resources it has. A warehouse at 82 percent on time shipment against a best in class 98 percent has a 16 point gap and might commit to 90 percent within a year. Targets set this way are grounded in evidence rather than in a round number chosen for comfort.

#75 · Organizational Process Management and Measures

What do revenue growth, market share, and margin each measure?

Answer

Revenue growth measures how much more the organization sells than in the prior period, market share measures its slice of the total market, and margin measures how much profit is left from each dollar of sales. A bakery with 20 percent revenue growth but a falling margin is selling more while keeping less, which points to pricing or cost problems. A Six Sigma project can move any of the three, so the team must name which one it targets.

#76 · Organizational Process Management and Measures

How is a finished project's financial benefit verified after the project closes?

Answer

Finance compares the post project result with the baseline using the same measurement definition that was agreed at the start. If a project claimed 200,000 dollars of annual savings from reduced rework, finance checks the rework account a quarter later to confirm the spending actually fell. Claims that cannot be traced to a ledger account are usually treated as soft benefits rather than verified savings.

#77 · Organizational Process Management and Measures

What is the customer retention rate, and how is it calculated?

Answer

Retention rate is the share of customers who stay over a period, usually calculated as the number retained divided by the number at the start. A subscription service that begins the year with 1,000 accounts and ends with 900 of them still active has a retention rate of 90 percent. Retention is a lagging measure of loyalty, so teams pair it with leading indicators such as complaint volume.

#78 · Organizational Process Management and Measures

How does a scorecard cascade from the enterprise level down to a process level?

Answer

Leaders start with the few enterprise measures that matter most, then translate each one into the value stream or department measure that drives it, and finally into the process measure a team can influence. A goal to raise on time delivery may become a warehouse measure for picking accuracy and then a process measure for label scan errors. The cascade works when a front line team can see how its measure moves the top level number.

#79 · Organizational Process Management and Measures

How is payback period calculated, and what does it miss?

Answer

Payback divides the project cost by the annual cash benefit to show how many years the project needs to return its money. A 150,000 dollar project that returns 50,000 dollars a year pays back in three years. Payback ignores everything after the break even point and ignores the time value of money, so a project with a long payback can still be a better investment than a quick one.

#80 · Organizational Process Management and Measures

How does customer satisfaction differ from customer loyalty as a performance measure?

Answer

Satisfaction measures how a customer feels about a recent experience, while loyalty measures whether the customer keeps buying and recommends the organization. A customer can report high satisfaction with a repair visit and still switch to a cheaper competitor at renewal. Teams track both because satisfaction alone does not prove that the relationship will continue.

#81 · Team Management

What team types can a Black Belt choose from?

Answer

A cross-functional team pulls members from several departments, a self-directed team manages its own work and decisions, and a virtual team works across locations using technology. The constraints of geography, technology availability, staff schedules, and time zones decide which type fits. A global supplier project usually runs as a virtual cross-functional team with members in three time zones.

#82 · Team Management

What are the standard team roles in a Six Sigma project?

Answer

The leader guides the team toward the goal and owns the results, the facilitator manages the process of the meeting and keeps the group on track, the coach helps members build skills, and each member contributes expertise and completes assigned work. One person can hold more than one role in a small team. A Black Belt often leads the project and facilitates the weekly meeting at the same time.

#83 · Team Management

What factors drive team member selection, and what makes a team succeed?

Answer

Selection looks at the ability to influence others, openness to change, needed skills, subject matter expertise, and availability. Success also needs management support, clear goals, ground rules, and timelines. A bank builds its loan project team by naming the underwriter, the systems analyst, and the branch supervisor who can actually change the workflow.

#84 · Team Management

What motivates a Six Sigma team, and what demotivates it?

Answer

Recognition, visible progress, meaningful goals, and a chance to learn motivate team members, while unclear roles, ignored input, repeated scope changes, and no credit for results demotivate them. A Black Belt can counter demotivation by restating the goal, giving feedback, and removing roadblocks. A team that sees its first improvement go live on the floor gains more energy than one that only hears about a future benefit.

#85 · Team Management

What are the classic stages of team development?

Answer

The stages are forming, storming, norming, performing, and adjourning. Forming is polite orientation, storming is conflict over roles and ideas, norming is agreement on how to work, performing is productive work, and adjourning is closing the team out. A Black Belt who expects storming and plans for it can keep a disagreement about measurement definitions from derailing the project.

#86 · Team Management

What belongs in a team communication plan?

Answer

A communication plan names the audience, the message type, the medium, and the frequency. It states who needs what information, in what form, and how often they receive it. A steering committee may get a one page monthly summary, while the project team gets a detailed weekly status update in a shared folder.

#87 · Team Management

Which leadership approach should a Black Belt use with a team?

Answer

The four approaches are direct, coach, support, and delegate, and the right one depends on how much skill and confidence the team has for the task. A new team needs direction, a struggling team needs coaching, a capable team needs support, and an expert team can be delegated to. A Black Belt might direct the team during its first data collection effort and delegate once the members run the measurement study on their own.

#88 · Team Management

How should a Black Belt handle groupthink and difficult group behaviors?

Answer

Techniques such as coaching, mentoring, and intervention help with dominant participants, reluctant participants, groupthink, rushing to finish, and digressions. Groupthink is when a group suppresses doubt in order to keep agreement. A facilitator can ask each member to write down an opinion before the discussion starts, which surfaces disagreement that a loud voice would otherwise bury.

#89 · Team Management

Which meeting and decision tools keep a project team moving?

Answer

Use an agenda, start on time, require pre-work before the meeting, and confirm the right people and resources are present. Then make decisions by consensus, by the nominal group technique, or by multi-voting, where the nominal group technique collects ideas silently before discussion and multi-voting gives each person a set number of votes. A team that argues for an hour over two options can multi-vote in five minutes and then discuss only the top choices.

#90 · Team Management

How is training for a Six Sigma team designed and evaluated?

Answer

Training starts with a needs assessment that finds the skills gap, then sets learning objectives, builds a training plan, and develops the materials. Delivery uses adult learning methods such as practice and real examples, and evaluation uses feedback surveys plus pre-training and post-training tests. A plant that trains operators on a new gauge checks whether they can read it correctly on production parts, not just whether they pass a written quiz.

#91 · Team Management

What are the steps of a training needs assessment?

Answer

The team identifies the skills gap against the work to be done, writes learning objectives, prepares a training plan, and develops the materials. A plant that books a statistics course before checking which Belts cannot read a control chart often trains the wrong people on the wrong content.

#92 · Team Management

How does a Black Belt tell whether a performance gap is a training problem or a process problem?

Answer

A training problem appears when people cannot perform the task even with the right tools, time, and information. A process problem appears when people know the task but the equipment, the layout, or the rules make it impossible, and sending those people to class wastes the training budget.

#93 · Team Management

Which modes of learning fit a team that has never used control charts?

Answer

A short explanation followed by hands-on practice with real data fits better than a lecture, because the skill is applied rather than recalled. Pairing a classroom session with floor coaching lets the team try the chart on live data while someone answers questions.

#94 · Team Management

What is the difference between a team charter and ground rules?

Answer

A team charter states the purpose, the scope, the goals, the roles, and the decision rights of the team. Ground rules govern how members behave in meetings and between them, such as starting on time, listening before responding, and raising problems to the team first.

#95 · Team Management

How does a facilitator handle a dominant talker and a silent member in one meeting?

Answer

The facilitator sets a speaking structure, such as a round robin or written ideas before discussion, so airtime is shared instead of seized. Directly inviting the quiet member's view after the dominant member speaks gives a specific opening without openly criticizing anyone.

#96 · Team Management

Why is rushing to finish a team's work dangerous?

Answer

Rushing pushes the team past the disagreement that would have surfaced a weak assumption, so the plan looks finished while the risk stays hidden. A team that skips the pilot to hit a date often spends more time fixing the rollout than the pilot would have cost.

#97 · Team Management

How does a Black Belt build trust with a team that has seen failed improvement projects before?

Answer

Trust is built by keeping small promises, reporting bad news early, and showing that the team acts on what members say. A Black Belt who announces a plan, asks for input, and then visibly changes the plan based on that input shows the team that this project will be different.

#98 · Team Management

What does a team leader do during the adjourning stage?

Answer

Adjourning is when the team finishes its work and disbands, so the leader closes out open items, captures lessons learned, and recognizes contributions. Members also need to know what happens next, such as who owns the process and how improvements will be reported.

#99 · Team Management

How does a Black Belt write a learning objective that can be measured?

Answer

A measurable objective names the action, the condition, and the standard, such as building an X-bar and R chart from a shift data set with limits correct to two decimals. If the objective cannot be observed or scored, the training cannot be evaluated.

#100 · Team Management

What makes a feedback survey useful for improving the next training round?

Answer

A useful survey asks about specific elements such as pace, practice time, examples, and materials, and it leaves room for a concrete suggestion. A single question asking whether the trainee liked the course produces a score that cannot guide any change.

#101 · Team Management

What are the main team types in Six Sigma?

Answer

A cross-functional team pulls members from different departments, a virtual team works across locations using technology, and a self-directed team manages much of its own work and decisions. Each type fits a different problem and a different set of constraints. A process that spans sales, production, and shipping usually needs a cross-functional team.

#102 · Team Management

What constraints drive the choice of team type?

Answer

Geography, time zones, technology access, staff schedules, and travel budget all shape whether a team can meet in person. A virtual team fits scattered experts, while a co-located team fits fast hands-on problem solving. The leader should match the team design to the real constraints instead of copying the last project.

#103 · Team Management

What factors guide team member selection?

Answer

A Black Belt looks for subject matter expertise, the skills the project needs, the ability to influence others, openness to change, and available time. A member who knows the process but cannot attend meetings slows the team, and a member with time but no knowledge adds little. The mix should cover the process, the customer, and the people who must carry out the change.

#104 · Team Management

What elements make a Six Sigma team successful?

Answer

Successful teams have management support, clear goals, agreed ground rules, a realistic timeline, and the right resources. Ground rules cover attendance, decision making, and how members treat each other. A team with a clear charter but no sponsor support often stalls when it needs money or access.

#105 · Team Management

What is the nominal group technique, and when does a team use it?

Answer

The nominal group technique has members write ideas silently, record each idea in turn, discuss the ideas only for clarity, and then rank them privately before the scores are combined. The silent steps keep a senior or talkative member from steering the group, and the private ranking gives every voice equal weight. A Black Belt may use it when a team must choose among five improvement options and one manager tends to dominate open discussion.

#106 · Team Management

How do the leader, facilitator, coach, and member roles differ?

Answer

The leader sets direction and owns results, the facilitator manages the meeting process and keeps discussion productive, the coach develops skills and thinking, and the member does the analysis and project work. One person can hold more than one role, but the team should know which hat is on. A leader who also facilitates must be careful not to dominate the discussion.

#107 · Team Management

What is a self-directed work team, and what must the organization provide for it?

Answer

A self-directed team plans its own work, assigns tasks among members, and makes day-to-day decisions without a manager directing each step. The organization still supplies a clear goal, boundaries, training, information, and access to resources, because self-direction without those supports turns into confusion. A production cell that sets its own job rotation and schedules its own maintenance is self-directed, yet it still needs targets and a budget from the plant.

#108 · Team Management

How does a consensus decision differ from a majority vote?

Answer

A majority vote lets just over half of the team decide and leaves the rest to carry out a choice they may oppose. Consensus means every member can support the decision and live with it even when it is not a first choice, so the team moves forward together. A Black Belt should not call a 6 to 5 vote a consensus when five members still object to the change.

#109 · Team Management

What leadership approaches fit different team situations?

Answer

Direct works when a member has low skill and low confidence, coach works when skill is low but willingness is high, support works when skill is high but confidence or motivation varies, and delegate works when skill and commitment are both high. The leader matches the approach to the person and the task, not to personal preference. A new analyst needs coaching, while a veteran process owner may need only delegation.

#110 · Team Management

What decision-making methods can a team use, and how does it handle groupthink?

Answer

Consensus means everyone can support the decision, nominal group technique has members write ideas silently before sharing and ranking them, and multi-voting narrows a list with weighted votes. Groupthink happens when pressure to agree stops people from raising doubts, so leaders can collect anonymous input and invite challenge. A team that rushes to agree may miss a serious risk that one quiet member noticed.

#111 · Team Management

Why does a project team need a defined escalation path?

Answer

When the team hits a barrier outside its authority, the sponsor must remove it quickly, and a defined path tells the team whom to ask and how fast to expect an answer. A team that waits three weeks for a budget decision loses the momentum and the calendar it built.

#112 · Team Management

Why should a project team stay small?

Answer

A small team can meet, decide, and act quickly, while a large team slows every decision and lets members hide, so teams often hold at five to seven core members. Advisors join only the topics they own, such as a maintenance planner for one session.

#113 · Team Management

What does an action log record, and why does the team review it first?

Answer

The action log records each task with one owner and a due date, and the team reviews it at the start of the next meeting so nothing is lost between sessions. An action without an owner is a wish, and it usually does not get done.

#114 · Team Management

What is a parking lot in a facilitated meeting?

Answer

A parking lot is a visible list where the facilitator holds items that matter but do not belong to the current agenda, so the group keeps the thought without losing the plan. A raised budget question goes to the parking lot with a named owner and a date to return.

#115 · Team Management

When should a Black Belt decide alone instead of asking the team?

Answer

A leader decides alone when the issue is urgent, the leader owns the outcome, or the team lacks the information to choose well, and the leader states why the call is being made. A safety stop should not wait for a vote, while a choice about the group's working hours belongs to the group.

#116 · Team Management

What should a team do when a member's availability drops mid-project?

Answer

The leader first renegotiates the workload with the member's manager, because the missing member takes process knowledge away with them. If the pull is permanent, the team replaces the member early and briefs the newcomer with the project charter and the data collected so far.

#117 · Team Management

What is a plus delta review at the end of a team meeting?

Answer

A plus delta review asks the team what worked well and what should change next time, and it takes only a few minutes at the close of the meeting. A team that agrees to send the data pack one day earlier is making a small delta decision that improves the next session.

#118 · Team Management

What should happen at a project kickoff meeting?

Answer

The kickoff sets the purpose, the scope, the roles, the ground rules, the timeline, and the first actions, and it gives every member a chance to raise a concern early. A kickoff that ends with named actions and dates starts the project with momentum instead of another meeting about meeting times.

#119 · Team Management

Why should the process owner stay involved while the project is still running?

Answer

The process owner holds the operating knowledge and will inherit the change, so early involvement prevents a solution that cannot be run and builds commitment to it. A team that designs a new check with no input from the supervisor may find that the shift cannot staff it.

#120 · Team Management

How does a decision log differ from an action log?

Answer

A decision log records what the team decided, when, and why, while an action log records who will do what by when. A decision to leave one supplier out of the pilot is written once in the decision log, and the phone calls that follow are actions.

#121 · Team Management

Why does a Six Sigma training plan include soft skills as well as technical tools?

Answer

Soft skills cover listening, giving feedback, running a meeting, resolving conflict, and influencing people who do not report to the belt. A belt who runs a flawless analysis but alienates the process owner will still lose the improvement at handoff, because the owner never accepted the change. Training plans that pair the statistics with practice in explaining results to a skeptical supervisor produce belts who can actually deploy the work.

#122 · Team Management

What does adult learning theory say about delivering project training?

Answer

Adults learn best when the material connects to a problem they already face, when they can practice it, and when they can bring their own experience into the session. A statistics class for belts works better when learners run the test on their own process data than when they watch a lecture on formulas. Short practice with immediate feedback beats a long slide deck.

#123 · Team Management

How is team training evaluated with pre training and post training testing?

Answer

The team measures skill before the training, delivers the training, and measures the same skill again afterward with an equivalent test. If agents scored 55 percent on a claim coding exercise before training and 85 percent after, the gain is evidence that the training worked. The evaluation should also check whether the skill still shows up in the work a month later, because a test score alone can fade.

#124 · Team Management

What ground rules should a Six Sigma team set at its first meeting?

Answer

Ground rules cover attendance, starting on time, how decisions get made, how members disagree, and how long a topic gets before it moves to a parking lot. A team might agree that every member speaks once before anyone speaks twice and that data resolve disputes rather than rank. The rules go into the team charter so a new member can see them in writing.

#125 · Team Management

What extra challenges does a virtual team face, and how are they handled?

Answer

A virtual team loses the casual conversation that builds trust and makes it hard to read whether a quiet member agrees or objects. Leaders handle this by setting a clear meeting rhythm, rotating the time so one region is not always inconvenienced, and using small breakout work instead of one long call. Written minutes and shared files replace hallway updates that remote members would otherwise miss.

#126 · Team Management

How does a facilitator manage a member who dominates the discussion?

Answer

The facilitator can use a structured round so each person speaks in turn, ask the dominant member to record ideas instead of talking, or use a silent brainstorm before discussion opens. A common move is to say that the team will hear three more views before returning to the first speaker. The aim is to keep the member's energy while making room for quieter voices.

#127 · Team Management

How does a facilitator draw out a reluctant participant?

Answer

The facilitator can ask a low risk question the person can answer from experience, use small pairs before the full group, or invite a written idea that gets read aloud without attribution. In a workshop, having each person write three causes on sticky notes before the group talks brings quiet knowledge into the room. The facilitator also watches for signals such as a change in expression that suggests the person disagrees and wants to speak.

#128 · Team Management

What is the risk when a team rushes to finish, and how is it managed?

Answer

A team that wants to be done can accept the first plausible cause and skip the verification step, which sends the project into a solution that does not hold. The leader slows the group by requiring evidence for the chosen cause and by running a small pilot before a full rollout. Naming the deadline pressure in the open helps the team separate speed of meeting from speed of the process.

#129 · Team Management

What happens during the adjourning stage of a team's life?

Answer

Adjourning is the stage where the project ends, the team disbands, and members move to new work. The leader captures lessons learned, confirms the handoff to the process owner, thanks the members for specific contributions, and releases them to other assignments. Teams that skip this stage lose the record of what worked and leave the process owner without a clear point of contact.

#130 · Team Management

How do coaching, mentoring, and intervention differ as ways to resolve team conflict?

Answer

Coaching helps a member build the skill or awareness to change their own behavior, mentoring pairs a less experienced member with a senior guide over a longer period, and intervention steps in directly to stop conduct that is harming the team. A Black Belt coaches a member who dominates meetings, mentors a Green Belt learning to run a chart, and intervenes when two members trade personal attacks. The choice depends on whether the problem is a skill gap, a development need, or active harm.

#131 · Define

How does a Black Belt identify and segment customers?

Answer

Customer identification lists everyone who receives or is affected by the process output, including internal customers such as the next department in line and external customers who buy the product. Segmentation groups customers by needs, volume, or value so the project can target the right group. A hospital lab has internal customers on the wards and external customers in clinics, and their turnaround needs differ.

#132 · Define

Which voice of the customer data collection methods should a project use?

Answer

Surveys reach many people at low cost, focus groups explore reactions in depth, interviews get detail from one person at a time, and observation shows what customers actually do rather than what they say. All four methods should be checked for validity, which means measuring the right thing, and reliability, which means giving consistent results. A software team that only surveys users may miss a problem that observation at the keyboard reveals in minutes.

#133 · Define

How do CTX and a CTQ tree turn a customer need into something measurable?

Answer

CTX stands for critical to X, where X can be quality, cost, safety, or delivery, and it names the category of need that matters. A critical-to-quality tree then starts with a broad need, breaks it into quality drivers, and turns each driver into a measurable requirement. A courier need for reliability might break into on-time departure and on-time arrival drivers, then into a requirement of 98 percent delivery within the promised window.

#134 · Define

What is quality function deployment?

Answer

Quality function deployment is a planning method that translates customer requirements, called the whats, into technical requirements, called the hows, in a matrix often called the house of quality. The matrix shows the strength of each relationship so engineers know which technical feature matters most. A team building an insulated cup can see that lid seal design drives both the leak complaint and the temperature retention requirement.

#135 · Define

What does a SIPOC diagram show?

Answer

A SIPOC diagram shows the suppliers, inputs, process, outputs, and customers of a process on one page. It sets the boundaries of the project before detailed mapping begins. At the start of a claims project, a SIPOC shows that the mailroom supplies claim forms, the process turns them into decisions, and the customer is the policyholder.

#136 · Define

What does the Kano model explain about customer satisfaction?

Answer

The Kano model sorts features into must-be requirements that cause dissatisfaction when missing, performance requirements where more is better, and attractive requirements that delight when present but do not upset when absent. Must-be needs are taken for granted once they are met. A clean room is a must-be, fast check-in is a performance feature, and a free room upgrade is an attractive feature.

#137 · Define

What belongs in a business case for a Six Sigma project?

Answer

A business case states the problem, the cost of the current performance, the expected benefit, the estimated cost and resources, the risk, and the link to strategy. It gives leaders the reason to fund the work. A business case for a billing project may show 400,000 dollars a year lost to rework against 90,000 dollars of project cost.

#138 · Define

What makes a good problem statement?

Answer

A good problem statement describes the pain in measurable terms, names where and when it occurs, states the baseline, and does not name a cause or a solution. It answers what, where, when, and how much. A weak statement says the shipping team is careless, while a strong one says 8 percent of east coast orders shipped late last quarter against a goal of 2 percent.

#139 · Define

How is project scope defined and reviewed?

Answer

Scope defines the boundaries of the project, including what is in scope, what is out of scope, the process start and stop points, and the affected locations or products. Clear boundaries keep the team from drifting and keep the project valuable to the customer. A billing project may include domestic invoices but list international invoices as out of scope.

#140 · Define

What makes a goal SMART?

Answer

SMART stands for specific, measurable, actionable, relevant, and timebound. A SMART goal states exactly what will change, how it will be measured, and by when. A goal to cut order entry errors from 6 percent to 2 percent by December 31 satisfies all five parts, while a goal to improve quality satisfies none.

#141 · Define

Which performance measurements connect a project to key outputs?

Answer

Project performance measures include cost, revenue, delivery, schedule, and customer satisfaction, chosen so they link critical process steps to the outputs that matter. The team also reviews the charter with stakeholders at planned checkpoints to confirm the goal still fits the business. A delivery project might track cost per shipment, on-time percentage, and customer complaint rate together.

#142 · Define

Which project management tools track a Six Sigma project?

Answer

A Gantt chart shows tasks as bars across a timeline, a work breakdown structure breaks the work into smaller deliverables, and toll-gate reviews are go or no-go checkpoints along the project. A RACI chart assigns who is responsible, accountable, consulted, and informed, and only one person should be accountable for a task even when several help. A RACI chart for a data collection step might list the Black Belt as accountable and the line supervisor as consulted.

#143 · Define

Which analytical tools are used across the DMAIC cycle?

Answer

An affinity diagram groups many ideas into themes, a tree diagram breaks a goal into levels of detail, a matrix diagram shows relationships between two lists, and a prioritization matrix ranks options by weighted criteria. An activity network diagram sequences tasks and shows the critical path, a process decision program chart maps what could go wrong and the countermeasure, and an interrelationship digraph shows cause and effect among many issues. A team drowning in 60 brainstormed ideas can use an affinity diagram to cut them to six themes before prioritization.

#144 · Define

What should an out-of-scope list include?

Answer

An out-of-scope list names the processes, plants, products, or customer groups the project will not change, and it can also name the solutions it will not consider. Writing the exclusions down at the start makes later requests visible as scope changes instead of silent additions.

#145 · Define

What makes a problem statement weak?

Answer

A weak problem statement blames a person, states a solution, or describes a feeling instead of a measurable gap. Saying the night shift is careless or that the plant needs a new machine closes the investigation before it starts, while naming the defect rate and the period keeps the cause open.

#146 · Define

Which letter of SMART is most often missing, and what does it cost?

Answer

The time-bound part is missing most often, because teams agree on what to improve but not on when it must be done. A goal to cut invoice errors from 6 percent to 2 percent is incomplete until it adds a date, and without the date no one can tell whether the project is behind.

#147 · Define

How do you test whether a CTQ is measurable?

Answer

A critical to quality characteristic is measurable when two people can look at the same unit and agree on the reading, the units are stated, and a target with a tolerance is given. If the requirement says the room should feel comfortable, it is still a voice of the customer statement and not yet a critical to quality characteristic.

#148 · Define

How do Kano categories change as a market matures?

Answer

A delighter in one generation becomes a must-be in the next, because customers fold today's surprise into tomorrow's expectation. A phone camera that once delighted buyers is now a feature whose absence causes complaints, so the team should refresh its Kano data instead of trusting an old survey.

#149 · Define

What does a SIPOC map not show?

Answer

A SIPOC map shows the suppliers, the inputs, the high-level process steps, the outputs, and the customers, but it does not show decisions, branches, time, or waste. A team that tries to diagnose a delay from a SIPOC alone needs a detailed process map instead.

#150 · Define

What does the roof of a house of quality reveal?

Answer

The roof shows where two technical characteristics pull against each other, such as lower weight fighting higher strength or a faster cure fighting a stronger bond. Recording the conflict early lets the team make the trade-off deliberately rather than discovering the clash during design.

#151 · Define

How does a team prioritize customer segments?

Answer

A team ranks segments by size, growth, profitability, and the value the organization can deliver to each one, then focuses the project where it can move a result. A segment that is large but already well served is usually a weaker target than a smaller segment with a clear unmet need.

#152 · Define

What does a RACI chart clarify when two people share a task?

Answer

RACI separates who does the work from who owns the result, so a task can have several responsible people but only one accountable person. When two people are marked accountable, a disagreement between them has no tiebreaker and the task stalls.

#153 · Define

How does a Gantt chart show dependencies?

Answer

A Gantt chart shows a bar for each task and arrows for the finish-to-start links that force one task to wait for another. The arrows expose the chain of waiting, so a slip on one bar shows immediately which later bars must move.

#154 · Define

What is the 100 percent rule in a work breakdown structure?

Answer

The 100 percent rule says the children of a work breakdown structure node must add up to exactly the parent, with nothing extra and nothing missing. If a deliverable appears at two levels or is dropped from the breakdown, estimates and assignments either double count the work or leave it unowned.

#155 · Define

What is a work package, and when is the breakdown deep enough?

Answer

A work package is the lowest level of the breakdown, small enough to estimate, schedule, assign, and control as one unit. The breakdown is deep enough when one owner can be held responsible for the package, and going deeper adds paperwork without adding control.

#156 · Define

How does a team connect a process measure to a key output?

Answer

The team starts from the key output the customer cares about, then traces backward to the process steps that drive it and picks a measure at one of those steps. A customer complaint about late deliveries might trace back to a measure such as the percent of orders released with a complete address.

#157 · Define

What is scope creep, and what signs show a project is drifting?

Answer

Scope creep is the slow addition of extra work, requirements, or customers to a project that was approved with a narrower charter. Warning signs include new deliverables that never appeared in the charter, a growing list of stakeholders asking for changes, and missed dates while the team works on tasks outside the original problem. A team asked to cut order entry errors has lost scope control when it starts redesigning the whole billing system.

#158 · Define

What voice of the customer collection methods are available?

Answer

Surveys reach many people with structured questions, focus groups produce rich discussion, interviews go deep with one person at a time, and observation shows what customers actually do rather than what they say. The method should match the question and the population. A team studying checkout waits may learn more from observation than from a survey.

#159 · Define

What is the difference between validity and reliability in customer data?

Answer

Validity means the method measures what it claims to measure, and reliability means it gives consistent results when repeated. A survey can be reliable but invalid if it consistently measures the wrong thing. A scale that always reads two pounds heavy is reliable but not accurate.

#160 · Define

What are CTX and CTQ?

Answer

Critical to X marks the qualities a customer cares about, where X can stand for quality, cost, safety, delivery, or another key need. Critical to quality translates the customer voice into a measurable requirement the project can track. A customer want for a quiet hotel room becomes a CTQ such as a sound level below a set number of decibels.

#161 · Define

What is a CTQ tree?

Answer

A CTQ tree starts with a broad customer need and breaks it into narrower needs and then into measurable requirements. Each branch must stay connected to the original voice of the customer. A need for fast service might branch into order accuracy and a delivery time under 24 hours.

#162 · Define

What is quality function deployment and the house of quality?

Answer

Quality function deployment is a structured way to turn customer requirements into technical design requirements. Its main matrix, called the house of quality, lists customer needs on one side, technical responses on another, and shows how strongly they connect. The roof of the house shows where technical responses support or conflict with each other.

#163 · Define

What is a SIPOC diagram?

Answer

A SIPOC diagram shows suppliers, inputs, process, outputs, and customers at a high level. It sets the process boundaries before a team maps the detailed steps. A billing team might list the sales system as a supplier, order data as an input, and the customer as the receiver of the invoice.

#164 · Define

What is the Kano model?

Answer

The Kano model sorts features into must-be requirements, performance requirements, and attractive or delighter features. Must-be features cause dissatisfaction when missing but do not raise satisfaction much when present, while delighters create surprise and pleasure. Reliable brakes are a must-be feature, and a free loaner car during service may be a delighter.

#165 · Define

What goes into a business case for a Six Sigma project?

Answer

A business case states the problem, the cost of the current state, the expected benefit, the risks, and the resources and timeline needed. It gives leaders a clear reason to fund the project and a way to compare it with other work. A weak business case promises savings without showing how they were estimated.

#166 · Define

What makes a strong problem statement?

Answer

A strong problem statement describes the gap between the current baseline and the desired goal using facts about what, where, when, and how much. It avoids naming a cause or a solution, because those come later in the project. A good example says that order errors rose from 2 percent to 5 percent since January, not that the team needs new software.

#167 · Define

How does a team set project scope?

Answer

Project scope defines the start and stop points of the process, what is inside the project, and what is explicitly outside. Clear boundaries protect customer value and stop the team from drifting into unrelated work. A team might include order entry and credit checks but exclude the warehouse picking process.

#168 · Define

What does SMART mean for project goals?

Answer

SMART stands for specific, measurable, actionable, relevant, and timebound. A SMART goal names the measure, the size of the change, the owner, and the deadline. Cutting invoice errors from 4 percent to 1 percent by December 31 is SMART, while improving billing is not.

#169 · Define

What is in a project charter, and which project management tools support it?

Answer

A project charter records the problem, goal, scope, business case, team, milestones, and measures, and it is reviewed with stakeholders at toll-gate points. A Gantt chart shows the schedule, a work breakdown structure divides the work into deliverables, and a RACI model assigns who is responsible, accountable, consulted, and informed. The charter is a living document, so the team updates it when the project changes.

#170 · Define

How does a milestone differ from a task in a project schedule?

Answer

A milestone is a zero-duration marker of a result, such as charter signed or pilot complete, while a task is work that takes time and consumes effort. A schedule shows the pilot as a two-week task and pilot complete as one milestone.

#171 · Define

What makes a survey question biased?

Answer

A question is biased when its wording leads the respondent or loads the answer, such as asking how much customers enjoyed the excellent new service. A neutral version asks how satisfied they were and lets the scale carry the judgment.

#172 · Define

Why should a focus group have a trained facilitator?

Answer

The facilitator keeps one voice from dominating, draws out quiet members, keeps the group on the topic, and prevents the group from settling before all views are heard. An untrained session often reports the views of the loudest two people.

#173 · Define

What are the limits of observation as a voice of the customer method?

Answer

Observation shows what people do but not why they do it, and people may change their behavior when they know they are watched. A team that sees customers abandon a form still needs an interview to learn what confused them.

#174 · Define

How does a Kano questionnaire classify a feature?

Answer

It asks the customer how they feel when the feature is present and again when it is absent, then crosses the two answers to place the feature in a category. A feature that pleases when present and bothers no one when absent is a delighter.

#175 · Define

How does a stakeholder differ from a customer?

Answer

A customer receives the output, while a stakeholder is anyone who affects or is affected by the process or the project, such as a regulator, a supplier, or a department that inherits the work. A payroll project serves employees as customers and treats the tax authority and the union as stakeholders.

#176 · Define

What is an interview guide, and how does a team build one?

Answer

An interview guide is a short list of open questions that keeps interviews consistent across customers without turning them into a survey. A team studying missed appointments asks the same five questions of each customer and records the answers in the customer's own words.

#177 · Define

How is a complaint log used as voice of the customer data?

Answer

A complaint log records each complaint with its date, type, and source, so the team can rank the causes and see whether the problem is growing or shrinking. A hotel that codes complaints by type finds that late check-in outranks every other cause in the summer.

#178 · Define

How do warranty claims act as customer feedback for a product?

Answer

Warranty claims show which failures customers actually experience, how often, and at what cost, and they arrive without asking anyone to fill out a survey. A toolmaker that sees one switch design behind 40 percent of claims has a customer-driven priority.

#179 · Define

How does a requirement differ from a specification?

Answer

A requirement states what the customer needs, and a specification states the measurable limit the design must meet to satisfy it. A customer requirement of quiet operation may become a specification of 45 decibels at one meter.

#180 · Define

How does a functional requirement differ from a nonfunctional requirement?

Answer

A functional requirement describes what the product or service must do, while a nonfunctional requirement describes how well it must do it, such as speed, reliability, safety, or usability. A kiosk must accept a payment, which is functional, and respond within 3 seconds, which is nonfunctional.

#181 · Define

What is a use case, and how does it capture a customer requirement?

Answer

A use case describes a goal a user has and the steps the system and the user take to reach it, including the exceptions that can interrupt the flow. A use case for returning an item covers the normal return, the missing receipt, and the expired window in one place.

#182 · Define

What does a customer journey map show that a process map does not?

Answer

A journey map follows the customer across departments and touchpoints, including the waits and handoffs the customer feels, while a process map follows the work inside one process. A mortgage journey shows the customer waiting a week between the appraisal and the approval, which no single department process owns.

#183 · Define

What is a toll gate review, and how does it differ from a status update?

Answer

A toll gate review is a scheduled go or no go decision at the end of a project phase, where the sponsor and stakeholders judge whether the work meets the exit criteria before the next phase starts. A status update reports progress, while a toll gate asks whether the project should continue, be changed, or be stopped. A failing gate can send the team back to redo the analysis or cancel the project before more money is spent.

#184 · Define

What does an affinity diagram do with a wall of brainstormed ideas?

Answer

An affinity diagram groups many brainstormed ideas into themes that the team names, so a long unorganized list becomes a small set of issues. After a workshop generates 60 causes of late shipments, the team might sort them into carrier issues, warehouse errors, and order entry mistakes. Seeing which theme holds the most cards helps the team decide where to look first.

#185 · Define

What is a tree diagram used for in a Define or Analyze project?

Answer

A tree diagram starts with one goal or problem and breaks it into smaller branches until the bottom level is something the team can act on. A goal to cut order errors might branch into errors by entry method, then by order type, then by the specific field that is mistyped. The diagram turns a broad aim into work packages that a team can assign.

#186 · Define

What does a matrix diagram show between two lists?

Answer

A matrix diagram places two lists on the rows and columns and marks where a relationship exists, often showing how strong each link is. A team might map customer requirements against process steps and find that one requirement is touched by six steps while another is touched by none. The empty cells and the crowded cells both carry information.

#187 · Define

What is an interrelationship digraph, and how does it separate drivers from outcomes?

Answer

An interrelationship digraph draws arrows between issues to show which one causes which, then counts arrows in and out for each issue. An issue with many arrows out and few in is a driver, while an issue with many in and few out is an outcome that will improve once the drivers are fixed. A team facing late reports might find that unclear ownership drives both rework and missed deadlines.

#188 · Define

How does an activity network diagram identify the critical path?

Answer

The diagram links tasks in their required order, estimates each duration, and calculates the longest path through the network. That longest path is the critical path, and any delay on it delays the whole project, while tasks with slack can slip without moving the finish date. A project with a three day delay on a critical task finishes three days late even if every other task meets its date.

#189 · Define

How does the voice of the customer differ from the voice of the process?

Answer

The voice of the customer states what the customer needs, while the voice of the process shows what the current process actually delivers. Customers who need a delivery within three days are giving the voice of the customer, while a process that averages five days is giving the voice of the process, and the gap between them becomes the project target. A team that listens only to the process can improve a number that no customer ever asked about.

#190 · Define

How does a team set criteria weights in a prioritization matrix?

Answer

The team agrees on the criteria, then divides 100 points or uses a rating scale to give each criterion a weight that reflects how much it matters. If cost matters twice as much as speed, cost might receive a weight of 0.40 and speed 0.20. The weighted scores then rank the options, and the weights are documented so reviewers can challenge them.

#191 · Define

Why does a project charter get reviewed with stakeholders during the project?

Answer

A periodic charter review checks whether the problem, scope, goal, and business case still match the process and the organization. Processes change, sponsors move, and a goal that made sense at the start can become irrelevant or already achieved. The review is also the moment to renegotiate scope before the team spends months on work the business no longer needs.

#192 · Define

How does a team establish the baseline performance for a problem statement?

Answer

The team defines the measure, the time period, and the population, then collects enough data to describe how the process performs now. A problem statement might open with a baseline of 12 late deliveries per 100 orders measured over the last quarter. Without a stated baseline and period, no one can prove later that the project changed anything.

#193 · Define

What is the difference between a customer need and a measurable customer requirement?

Answer

A need is what the customer wants in their own words, such as a repair that is done quickly, while a requirement states the need as something the process can be measured against. The need for a fast repair becomes a requirement that 95 percent of repairs are completed within 24 hours. Translating the need into a number is what lets the team test a solution.

#194 · Define

What does critical to X mean when X is safety or delivery rather than quality?

Answer

Critical to X names the attribute the customer cares about most, so critical to quality covers defects, critical to safety covers harm, and critical to delivery covers timeliness. A courier's customer may rank critical to delivery above critical to quality, which shifts the project toward on time arrival. The X is chosen from customer data rather than from what the engineering team finds easiest to measure.

#195 · Define

What is the difference between a milestone and a deliverable in a project plan?

Answer

A deliverable is a tangible output the team produces, such as a validated measurement system or a pilot report, while a milestone is a point in time that marks progress, such as the date the charter is approved. A Gantt chart shows milestones as markers and deliverables as work that must be completed to reach them. Confusing the two makes a team report activity as if it were a result.

#196 · Measure

Which process flow metrics does a Black Belt use to find constraints?

Answer

Work in progress is the amount of work sitting in the process, touch time is the time actually worked on a unit, cycle time is the elapsed time from start to finish for one unit, throughput is the number of units finished per period, and takt time is the pace the customer demands. A plant with 450 minutes of available work time and demand of 900 units has a takt time of 450 divided by 900, or 0.5 minutes (30 seconds) per unit. When the cycle time is longer than the takt time, the process cannot keep up and the constraint becomes visible.

#197 · Measure

What is a hidden factory?

Answer

A hidden factory is the extra work inside a process that the official process map does not show, such as rework, sorting, expediting, and repeat inspections. It raises the real cost and the real cycle time while the documented process looks clean. An order desk whose map shows five steps may actually re-key 40 percent of orders after errors, which adds labor that no one tracked.

#198 · Measure

Which process analysis tools should a Black Belt select and evaluate?

Answer

Value stream maps show the flow of material and information with value added and waiting time, process maps show the steps of a process, flowcharts show decision paths, spaghetti diagrams trace the physical movement of people or parts, work instructions document how a task is done, and a gemba walk means going to where the work happens to see it directly. The tool is chosen to match the question being asked. A team studying excessive walking uses a spaghetti diagram, while a team studying delay uses a value stream map.

#199 · Measure

What is the difference between qualitative and quantitative data, and between continuous and discrete data?

Answer

Qualitative data describes categories or opinions, such as complaint type, while quantitative data is numeric, such as weight or time. Continuous data can take any value in a range, such as 12.43 mm, while discrete data comes in counts or categories, such as the number of defects. The data type decides which chart and which statistical test are valid.

#200 · Measure

What are the four measurement scales?

Answer

Nominal data only names or labels, such as part number or machine ID, with no order. Ordinal data has a meaningful order but unequal or unknown spacing, such as a 1 to 5 satisfaction rating, interval data has equal spacing but no true zero, such as temperature in degrees Celsius, and ratio data has equal spacing with a true zero, such as weight, time, or dollars. Only ratio data supports statements such as twice as much.

#201 • Measure

Which sampling methods and data collection pitfalls should a Black Belt watch?

Answer

Random sampling gives every unit an equal chance, stratified sampling splits the population into groups and samples inside each, systematic sampling takes every kth unit, subgroup sampling collects consecutive units at planned intervals for control charts, and block sampling groups similar units and samples within the block. A sample must be representative, homogeneous, accurate, and precise, with no bias. Data collection plans fail when operational definitions are vague, collectors are untrained, too much data is gathered, or seasonality is ignored.

#202 • Measure

What does a measurement system analysis cover?

Answer

A gauge repeatability and reproducibility study separates repeatability, the variation when one operator measures the same part several times, from reproducibility, the variation between operators. Bias is the average difference from a true reference, linearity checks that bias across the measurement range, stability checks consistency over time, and precision to tolerance compares gauge variation with the tolerance width, while attribute systems use percent agreement and audit studies with known parts. Metrology adds calibration, traceability to national standards, and control of the measuring devices themselves.

#203 · Measure

How does a population parameter differ from a sample statistic?

Answer

A population parameter is the true value for the whole group, such as the population mean or the population standard deviation, and it is usually unknown. A sample statistic is the value calculated from a sample, such as the sample mean, the sample standard deviation, or a sample proportion, and it estimates the parameter. A study that describes only its own sample is descriptive, while one that uses the sample to draw a conclusion about the population is inferential.

#204 · Measure

What does the central limit theorem say and why does it matter?

Answer

The central limit theorem says that the distribution of sample means approaches a normal distribution as the sample size grows, even when the underlying data is not normal. This holds for most distributions once the sample size reaches about 30. The theorem is what makes confidence intervals, hypothesis tests, and control charts work on real process data.

#205 • Measure

Which descriptive statistics and graphs summarize a data set?

Answer

The mean, median, and mode describe the center, while the range, variance, standard deviation, and interquartile range describe the spread. A histogram shows shape and spread, a box-and-whisker plot shows the median, quartiles, and outliers, a scatter diagram shows the relationship between two variables, a normal probability plot checks normality, and frequency and cumulative frequency distributions show counts and running totals. A skewed data set is better described by its median than by its mean.

#206 • Measure

What are the basic rules of probability?

Answer

Mutually exclusive events cannot happen at the same time, so the addition rule gives the probability of either event as the sum of their probabilities. Independent events do not affect each other, so the multiplication rule gives the probability of both as the product of their probabilities. Conditional probability is the chance of one event given that another happened, and the complement rule says the chance of an event not happening is one minus the chance that it does.

#207 · Measure

When is the normal, binomial, or Poisson distribution the right model?

Answer

The normal distribution models continuous measurements that cluster around a mean, such as fill weight or cycle time. The binomial distribution models the number of successes in a fixed number of independent trials with a constant success chance, such as the number of defective parts in a sample of 50. The Poisson distribution models the count of rare events in a unit of time or space, such as the number of customer complaints per day.

#208 · Measure

Which other distributions should a Black Belt recognize?

Answer

The Student t distribution is used for means when the population standard deviation is unknown and has heavier tails than the normal, the chi square distribution is used for variances and count tests, and the F distribution compares variances and appears in analysis of variance. The hypergeometric distribution covers sampling without replacement from a small finite lot, the exponential distribution models time between events, the lognormal distribution fits right skewed positive data, and the Weibull distribution is widely used for life and reliability data. Each one matches a different shape of real process data.

#209 · Measure

What do Cp and Cpk measure?

Answer

Cp compares the specification width with six standard deviations of the process, so it shows potential capability but ignores centering. Cpk compares the distance from the process mean to the nearer specification limit with three standard deviations, so it shows actual capability. With limits of 10 and 20, a mean of 17, and a standard deviation of 1, Cp is 1.67 while Cpk is only 1.00 because the process is off center.

#210 · Measure

How do Pp, Ppk, Cpm, and process sigma differ from Cp and Cpk?

Answer

Pp and Ppk use the overall standard deviation of all the data, while Cp and Cpk use the within-subgroup standard deviation, so Pp and Ppk reflect both common and special cause variation. Cpm also compares the process mean with the target value, so it drops when the process is off target even if it sits inside the limits. Process sigma converts a defect rate into a sigma level using the 1.5 sigma shift convention, where 3.4 defects per million opportunities equals six sigma.

#211 • Measure

Which performance metrics describe process output, and how is rolled throughput yield calculated?

Answer

Defects per unit is defects divided by units, defects per million opportunities scales defects by the number of chances for error, and parts per million is the count of defective units scaled to one million parts. First pass yield is the share of units that pass without rework, and rolled throughput yield multiplies the first pass yields of all the steps. A three step process with yields of 95, 90, and 98 percent has a rolled throughput yield of 0.95 times 0.90 times 0.98, or about 0.838, meaning 83.8 percent, and attributes capability converts these counts into a sigma level using the same defect table used for continuous data.

#212 • Measure

When does a capability study need special handling?

Answer

A capability study needs a stable process, a measurement system that is good enough, a sample that represents the process, and a check that the data is normal before C_p and C_{pk} are calculated. Non-normal data can be handled with a transformation such as Box-Cox or with a capability method fitted to the right distribution. Running a capability study on an unstable process produces numbers that will change tomorrow.

#213 · Measure

What is the difference between a descriptive and an inferential statistical study?

Answer

A descriptive study summarizes the data that were collected, such as the mean, range, or defect rate of a sample. An inferential study uses the sample to make a claim about a larger population, such as a confidence interval or a hypothesis test.

#214 · Measure

What is a sampling frame, and why does a missing frame bias a sample?

Answer

A sampling frame is the list or source the sample is actually drawn from, such as a customer database or a shift log. If the frame leaves out part of the population, such as cash customers missing from the database, the sample cannot represent the whole and the estimate will be biased.

#215 • Measure

What does a normal probability plot reveal that a histogram cannot?

Answer

A normal probability plot compares each data value with the value a normal distribution would predict, and a straight line means the data fit a normal model. A histogram can look roughly bell shaped while the plot shows curved tails, which matters because many capability calculations assume normality.

#216 • Measure

How is the sample standard deviation calculated?

Answer

Subtract the mean from each value, square each difference, add the squares, divide by one less than the number of values, and take the square root. For 4, 6, 6, 7, and 7, the mean is 6, the squared differences are 4, 0, 0, 1, and 1, the variance is 6 divided by 4, which is 1.5, and the standard deviation is about 1.22.

#217 · Measure

What is the danger of measuring only what is easy to measure?

Answer

Teams drift toward the measure that is easy to collect, such as the number of calls handled, even when the customer cares about resolution on the first call. The easy measure improves while real performance does not, which is why the team starts from the customer requirement and then finds a way to measure it.

#218 · Measure

What is the interquartile range, and why is it resistant to outliers?

Answer

The interquartile range is the third quartile minus the first quartile, so it covers the middle half of the data. Because it ignores the extreme values at each end, one wild reading changes the interquartile range very little while it can move the standard deviation a great deal.

#219 · Measure

What is the difference between stratified sampling and cluster sampling?

Answer

Stratified sampling divides the population into groups and draws from each group in proportion, which tightens the estimate of the whole. Cluster sampling selects whole groups, such as entire stores or shifts, and studies everything inside them, which is cheaper when travel or setup is the main cost.

#220 · Measure

What is an operational definition, and why does a metric need one?

Answer

An operational definition states exactly what is measured, the instrument, the procedure, and the pass or fail rule, so two people get the same result. Without one, an inspector who measures a scratch at an angle and one who measures it straight on record different defect rates without the process changing.

#221 · Measure

How can measurement systems be applied outside manufacturing?

Answer

Every function that produces a number has a measurement system, such as survey wording in marketing, code review in engineering, or order accuracy in customer service. A call center that never checks whether two reviewers score the same call the same way is running an unstudied measurement system.

#222 · Measure

What is the difference between random error and systematic error?

Answer

Random error scatters readings around the true value and shrinks when readings are averaged, while systematic error shifts every reading the same way and does not average out. A gauge that reads two grams high every time has systematic error, so averaging a thousand readings still gives an answer that is two grams wrong.

#223 · Measure

Why does the bin width change the shape of a histogram?

Answer

The bin width controls how much detail the histogram shows, and a width that is too large can hide a second peak while one that is too small turns random noise into false gaps. Trying two or three widths before choosing one keeps the shape of the distribution from being an artifact of the setting.

#224 · Measure

How is a sigma level calculated from attribute data?

Answer

For attribute data, defects per million opportunities are converted to a sigma level using the normal distribution, with the standard 1.5 sigma shift built into the convention. A process at 3,000 DPMO sits near 4.25 sigma on that convention, while 3.4 DPMO is called six sigma.

#225 • Measure

How do natural process limits differ from specification limits?

Answer

Natural process limits describe what the process actually produces, usually the mean plus and minus three standard deviations. Specification limits come from the customer or the design, so a process can sit comfortably inside its natural limits and still be unable to meet the specification.

#226 • Measure

How does conditional probability change a defect investigation?

Answer

Conditional probability is the chance of one event given that another event has already happened, written as the probability of A given B. If a defect appears only after a certain machine, the probability of a defect given that machine is far higher than the overall defect rate, which points the investigation at that machine.

#227 · Measure

What is homogeneity, and why does it matter in sampling?

Answer

Homogeneity means the items inside a sample or subgroup are similar enough that the variation between them reflects only normal process noise. Drawing a sample that mixes two different suppliers hides the difference between them, which is why a rational subgroup stays within one stream.

#228 · Measure

What is data integrity, and what threatens it in a collection plan?

Answer

Data integrity means the records are complete, accurate, and traceable to the point where they were captured. Transcribed numbers, missing rows, and two systems that record the same event with different timestamps all threaten integrity, so the team should trace a small batch by hand before trusting the file.

#229 · Measure

What is the difference between internal validity and external validity?

Answer

Internal validity means the study really shows the effect of the factor inside the study, without a lurking variable explaining it away. External validity means the result still holds in other lines, shifts, or plants, and a tight laboratory study can have strong internal validity while failing the moment it moves to the floor.

#230 · Measure

What are WIP and WIQ?

Answer

Work in progress is the amount of work that has started but is not finished, and work in queue is the work waiting for the next step. High levels of either one lengthen lead time and hide problems. A claims office with 400 open files and 90 waiting for review has both high work in progress and a long queue.

#231 · Measure

How is takt time calculated?

Answer

Takt time equals available work time divided by customer demand, and it sets the pace the process must match. With 480 minutes available and demand of 240 units, takt time is 2 minutes per unit. A cycle time above takt time means the process cannot keep up without overtime or a second resource.

#232 · Measure

What is the difference between cycle time and lead time?

Answer

Cycle time is the time to complete one unit or one step, while lead time is the total time from customer request to delivery. Lead time includes waiting time between steps, so it is usually longer than cycle time. A part may take 5 minutes of cycle time at a machine but 12 days of lead time through the plant.

#233 · Measure

What are throughput and touch time?

Answer

Throughput is the number of units the process completes in a given period, and touch time is the actual work time spent on one unit. The value-added ratio divides touch time by lead time to show how much of the total time adds value. If touch time is 12 minutes and lead time is 60 minutes, the value-added ratio is 12 divided by 60, or 20 percent.

#234 · Measure

What is a swimlane diagram, and what does it show that a simple process map does not?

Answer

A swimlane diagram places each step in a lane that belongs to one person, department, or system, so handoffs and waiting between owners become visible. A simple process map shows the order of steps but can hide how many times work crosses between departments. Mapping an order with lanes for sales, credit, and shipping often shows that the order waits three separate times for a signature.

#235 · Measure

How does a value stream map differ from a process map?

Answer

A process map shows the steps, decisions, and handoffs in a process, while a value stream map also shows information flow and the time spent waiting at each step. The value stream view separates value-added from non-value-added time. A team can see that only 15 minutes of a 5-day approval path is real work.

#236 · Measure

What are a spaghetti diagram and a gemba walk?

Answer

A spaghetti diagram traces the physical path that people or materials travel, which exposes wasted motion and transport. A gemba walk means going to the actual place where work happens to observe the real process. A nurse who walks 6 miles in a shift shows up clearly on a spaghetti diagram.

#237 · Measure

What are the main types of data in a Six Sigma project?

Answer

Qualitative data describes qualities or categories, while quantitative data uses numbers. Continuous data come from measurement, such as 10.2 millimeters, and discrete data come from counting, such as 3 defects. A gauge that reads 10.2 produces continuous data, while a count of rejected parts produces discrete data.

#238 · Measure

What five values does a box plot display, and what does the box width show?

Answer

A box plot shows the minimum, the first quartile, the median, the third quartile, and the maximum, with the box drawn from the first quartile to the third. The box width is the interquartile range, which holds the middle 50 percent of the values, and points beyond the whiskers are flagged as possible outliers. A box plot of hold times with a box from 10 to 16 minutes and one point at 31 minutes shows a tight middle half and a single unusually slow batch.

#239 · Measure

What are the common sampling methods?

Answer

Random sampling gives every unit the same chance of selection, stratified sampling divides the population into groups and samples each, and systematic sampling takes every nth unit. Subgroup sampling collects rational subgroups over time, and block sampling groups similar conditions to reduce noise. A lab that samples one batch from each supplier uses stratified sampling.

#240 · Measure

What do bias, accuracy, and precision mean in measurement?

Answer

Bias is a systematic error that pushes readings away from the true value in one direction, accuracy is closeness to the true value, and precision is closeness of repeated readings to each other. A scale can be precise but biased if it always reads the same amount too high. MSA studies test all three so decisions rest on trustworthy numbers.

#241 · Measure

What belongs in a data collection plan?

Answer

A data collection plan defines the metric, the operational definition, who collects the data, what to record, when and where to record it, and how the data will be checked. Teams use check sheets and train collectors so everyone measures the same way. A plan should also watch for seasonality, such as higher call volume every Monday.

#242 · Measure

What is a gauge R and R study, and why does calibration matter?

Answer

A gauge repeatability and reproducibility study separates variation from repeated readings by one operator, called repeatability, from variation between operators, called reproducibility. The percent gauge R and R compares measurement variation with total variation, and a high value means the gauge cannot support the decision. Calibration and traceability connect the gauge back to a known reference standard so readings stay valid over time.

#243 · Measure

What are the main descriptive statistics for center and spread?

Answer

Measures of center include the mean, median, and mode, and measures of spread include the range, variance, and standard deviation. The mean of 10, 12, and 14 is 12, and the standard deviation shows how far values typically sit from that mean. The median is useful when a few extreme values would pull the mean away.

#244 · Measure

What does the central limit theorem say, and what is the normal distribution?

Answer

The central limit theorem says that the distribution of sample means approaches a normal distribution as sample size grows, even when the underlying population is not normal. The normal distribution is a symmetric bell curve where the mean, median, and mode are equal. About 68 percent of values fall within one standard deviation of the mean, about 95 percent within two, and about 99.7 percent within three.

#245 • Measure

What is the difference between a sample and a census, and when is each used?

Answer

A sample measures part of the population, while a census measures every unit in it. A census removes sampling error but usually costs more and takes longer, and it can still carry measurement error, so teams sample when the population is large or the test is destructive. A plant that checks 20 of 500 bearings in a day is sampling, while checking all 500 would destroy the whole lot.

#246 • Measure

What are DPMO, DPU, PPM, first pass yield, and rolled throughput yield?

Answer

DPMO is defects per million opportunities, DPU is defects per unit, and PPM is defective parts per million. First pass yield is the share of units that pass a step without rework, and rolled throughput yield multiplies the first pass yields of all steps together. Twelve defects in 500 units with 4 opportunities per unit gives 12 divided by 2,000, or 6,000 DPMO, and steps at 95, 90, and 98 percent first pass yield give a rolled throughput yield of 83.8 percent.

#247 · Measure

Why does the standard deviation square the deviations instead of using absolute values?

Answer

Squaring makes every deviation positive and gives extra weight to large misses, and the result has useful mathematical properties for later tests. The absolute value version, called the mean absolute deviation, is easier to explain but less useful in the formulas that follow.

#248 · Measure

Why do Six Sigma tables apply a 1.5 sigma shift?

Answer

Short-term studies understate the drift that a process shows over months, so the convention shifts the mean by 1.5 standard deviations to reflect long-term performance. A process with 3.4 defects per million opportunities in the long run is called six sigma rather than 4.5 sigma.

#249 · Measure

When does a team choose a variable gauge study instead of an attribute agreement study?

Answer

It studies a variable gauge when the measurement is a number on a continuous scale, and it studies attribute agreement when the call is pass or fail, good or bad, or a category. A team that records a diameter runs a gauge R&R, and a team that records a visual pass or fail runs an attribute study.

#250 · Measure

What is the difference between a defect and an opportunity in DPMO counting?

Answer

An opportunity is a chance for the unit to fail a requirement, and a defect is one actual failure of a requirement. A form with 20 fields has 20 opportunities, so one form with 3 wrong fields carries 3 defects on 1 unit.

#251 · Measure

How does the opportunity count change the DPMO for the same process?

Answer

DPMO divides the defects by the total opportunities, so adding opportunities lowers the rate for the same defects. Ten defects across 500 units gives 5,000 defects per million opportunities when each unit has 4 chances and 4,000 when each unit has 5 chances.

#252 · Measure

What is the difference between the resolution of a gauge and its accuracy?

Answer

Resolution is the smallest change the gauge can display, and accuracy is how close the reading is to the true value. A scale that reads to 0.001 g but sits 0.5 g off the true value has fine resolution and poor accuracy.

#253 · Measure

When is the range a useful measure of spread?

Answer

The range is the largest value minus the smallest, and it is quick to compute for a small sample, which is why it drives the R chart. With five readings of 12, 14, 15, 15, and 18 minutes, the range is 6 minutes.

#254 · Measure

What does a dot plot show that a summary number hides?

Answer

A dot plot stacks one dot for each reading along a scale, so the shape, the gaps, and the outliers are visible at a glance. Twenty cycle times plotted this way may show two clusters that the average alone would hide.

#255 • Measure

How is a standard time built for a work element?

Answer

A standard time adds allowances to the normal time, where the normal time is the observed time adjusted for the worker pace, and allowances cover personal needs, fatigue, and delays. A task that runs 5.0 minutes at a normal pace with a 12 percent allowance has a standard time of 5.6 minutes.

#256 • Measure

What is a defect concentration diagram?

Answer

A defect concentration diagram is a picture of the product or form with a mark where each defect was found, so the marks show whether defects cluster in one area. A mark map showing most scratches near one corner points to a handling problem rather than a material problem.

#257 · Measure

What does a scatter diagram show before any regression is fitted?

Answer

The scatter diagram shows the direction, the shape, and the strength of the relationship and reveals outliers before any line is fitted. A team that plots the data first can see a curved pattern that a straight-line regression would describe poorly.

#258 · Measure

How does a histogram differ from a bar chart?

Answer

A histogram groups continuous measurements into bins that touch, so the shape of the distribution is visible, while a bar chart compares separate categories with gaps between the bars. Fill weights belong in a histogram, and defect types belong in a bar chart.

#259 · Measure

How does the observer effect distort a time study?

Answer

People work differently when someone is timing them, so the recorded times may be faster or more careful than normal work. The observer reduces the effect by staying unobtrusive, explaining the purpose, and taking readings across several days.

#260 · Measure

How does work sampling estimate how time is spent?

Answer

Work sampling takes many random observations and counts how often the worker or machine is in each state, so the percentages estimate the time split without a stopwatch. Two hundred random observations showing 30 percent waiting suggest the wait is real, but they show nothing about why it happens.

#261 • Measure

What does queuing theory say happens to waiting as a process approaches full use?

Answer

Waiting time grows slowly at low use and rises steeply as the process approaches full capacity, because the same variation has less slack to absorb it. A department running at 95 percent use can add a small amount of work and see the queue double.

#262 • Measure

What is a pass or fail standard in an attribute measurement system?

Answer

It is the written rule and physical reference that tells the appraiser where the boundary between pass and fail sits, including the doubtful zone. Two inspectors disagree on 30 percent of parts when no standard defines the boundary, and they agree once a limit sample and a rule are in place.

#263 · Measure

How should a data collection plan handle missing values?

Answer

The plan states up front how a missing value will be recorded, how often one is expected, and what will be done about it, and the collector writes the reason next to the gap instead of leaving a blank. A team that silently drops the days when the line was down will overstate the performance of the process.

#264 · Measure

What is linearity in a measurement system, and how is it tested?

Answer

Linearity is how consistent a gauge's error is across the range of parts it measures, because a gauge can read accurately at one end and poorly at the other. A scale that reads 0.2 grams low for a 10 gram part and 1.5 grams high for a 100 gram part has a linearity problem even if its average error looks small. The test compares the gauge's bias at several known reference values across the range.

#265 · Measure

What is stability of a measurement system?

Answer

Stability is how well a gauge holds its accuracy over time when it repeatedly measures the same master part. A gauge that reads a master correctly in January and 1.0 unit high in April has drifted and needs calibration or repair. Teams track stability with a control chart on a standard part, so drift shows up before it corrupts months of process data.

#266 · Measure

What is the precision to tolerance ratio, and what value signals trouble?

Answer

The precision to tolerance ratio multiplies the gauge standard deviation by 6 and divides by the tolerance width, which shows how much of the specification the gauge's own variation consumes. A gauge with a standard deviation of 0.02 against a tolerance of 0.20 gives a ratio of 0.12 divided by 0.20, or 60 percent, which is far too high. Many teams treat a ratio below 10 percent as good, 10 to 30 percent as marginal, and above 30 percent as unacceptable.

#267 · Measure

What is metrology, and what do calibration and traceability require?

Answer

Metrology is the science of measurement, covering the devices, standards, procedures, and environment used to produce a number. Calibration compares a device against a standard of known value, and traceability means that standard can be linked back through an unbroken chain to a national reference. A torque wrench is calibrated against a master that is itself traceable, so its readings can be defended in an audit.

#268 · Measure

What is a check sheet, and how does it help a data collection plan?

Answer

A check sheet is a simple form on which data are recorded as marks or tallies while the work happens. A team counting the reasons for a rework loop might draw a sheet with a column for each reason and tally a mark every time one occurs, which makes the pattern visible in a day. The sheet also enforces a shared definition, because the collector must choose a single category for each event.

#269 · Measure

What is data normalization, and when does it belong in a collection plan?

Answer

Data normalization means putting data from different sources into one agreed definition, unit, and format before analysis. If three plants each count a defect differently, a combined defect rate is meaningless until the team rewrites the definition and recodes the history. The rule is to normalize before comparing, never to compare raw numbers that were collected under different rules.

#270 · Measure

What is a seasonality effect, and how does a team keep it from distorting a study?

Answer

A seasonality effect is a repeating pattern tied to the calendar, such as a spike in orders every December or a slowdown every summer. A team that measures a process only during its busiest month will overstate the normal workload and misread the baseline. The plan avoids this by collecting across a full cycle or by comparing the same months in different periods.

#271 • Measure

How does a flowchart differ from a work instruction?

Answer

A flowchart shows the steps of a process and the decisions that route work between them, while a work instruction tells one person exactly how to perform a single step. A flowchart might show that a loan application goes to underwriting only when the amount exceeds 50,000 dollars, while the work instruction explains how to enter the underwriting request. Teams use the flowchart to find the problem and the work instruction to hold the fix.

#272 • Measure

What makes a sample representative, and what does homogeneity mean?

Answer

A representative sample has the same mix of conditions as the population, so conclusions from it apply to the whole process. If a plant runs three shifts and the team samples only day shift, the sample is not representative, because night shift conditions are missing. Homogeneity means the units within a subgroup are similar enough to treat as one snapshot, which is why a subgroup is usually taken close together in time.

#273 · Measure

How does sampling bias differ from measurement bias?

Answer

Sampling bias comes from which items are selected, such as surveying only customers who use the website when 40 percent of orders arrive by phone. Measurement bias comes from the gauge or the observer, such as a scale that reads 0.5 kilograms high for every part. One is fixed by changing the selection method, and the other is fixed by calibrating or replacing the measurement system.

#274 · Measure

How is a process sigma level estimated from DPMO?

Answer

The team converts defects per million opportunities into an area under the normal curve, then reads the corresponding z value and adds the 1.5 sigma shift convention. A process at 6,210 DPMO sits near four sigma, 233 DPMO near five sigma, and 3.4 DPMO at six sigma. The shift convention exists because processes drift over the long term, so a short term capability looks better than the long term result.

#275 · Measure

What is percent defective, and how does it relate to parts per million?

Answer

Percent defective is the share of units that fail to meet any requirement, and parts per million restates that share per million units. Twelve defective units out of 400 is 3 percent defective, which is 30,000 parts per million. Percent defective counts units, so it answers a different question from DPMO, which counts defects against opportunities.

#276 · Measure

What is the difference between a defect and a defective unit?

Answer

A defect is a single failure to meet a requirement, while a defective unit is a unit that has at least one defect. One shirt with three loose buttons has three defects but is one defective shirt. Teams use defects per unit to describe the density of problems and percent defective to describe how many units a customer would reject.

#277 · Measure

How do short term and long term capability differ?

Answer

Short term capability uses a standard deviation from a stable period of data, while long term capability uses data across weeks or months that include more sources of shift and drift. The long term standard deviation is usually larger, so Ppk comes out lower than Cpk on the same process. The 1.5 sigma shift convention exists to account for that gap when only short term data are available.

#278 · Measure

What is a cumulative frequency distribution, and what does it show?

Answer

A cumulative frequency distribution adds each interval's count to the running total, so the last point always reaches the full sample size. If 25 orders took under two days, 45 took under four days, 70 took under six days, and 90 took under eight days, the curve shows that 90 of 100 orders finished within eight days. Teams use it to answer questions such as what share of orders finish within a promised time.

#279 · Measure

How does sampling error differ from non sampling error?

Answer

Sampling error is the difference between a sample result and the true population value that comes from measuring only part of the population, and it shrinks as the sample grows. Non sampling error comes from mistakes such as a biased selection method, a misread gauge, or a data entry slip, and a larger sample does not fix it. A sample mean of 41.2 against a true mean of 40.5 has a sampling error of 0.7, but a scale that reads high every time is non sampling error.

#280 · Measure

How is conditional probability used to revise the chance of an event?

Answer

Conditional probability is the chance of one event given that another has already happened, calculated as the joint chance divided by the chance of the condition. If 30 percent of orders come from supplier A and 0.6 percent of all orders are both from A and defective, then the defect chance given supplier A is 0.006 divided by 0.30, which is 2 percent. This is how a team separates a supplier's defect rate from its share of the total volume.

#281 · Analyze

What does the correlation coefficient tell you, and why is correlation not causation?

Answer

The correlation coefficient ranges from minus 1 to plus 1 and measures the strength and direction of a linear relationship between two variables, with 0 meaning no linear relationship. A strong correlation does not prove that one variable causes the other, because both may be driven by a third variable. Ice cream sales and drowning deaths correlate because both rise in hot weather, not because ice cream causes drowning.

#282 · Analyze

How is a simple linear regression model built and checked?

Answer

A least squares regression line minimizes the sum of the squared vertical distances from the data points to the line, and it is written as y equals b_0 plus b_1 times x , where b_1 is the slope and b_0 is the intercept. R squared is the share of variation in y explained by x , and it equals the correlation squared. The model is checked by plotting the residuals to confirm they are random, roughly normal, and evenly spread, because a curve or a fan shape means the model is wrong.

#283 · Analyze

What is multiple regression and how does adjusted R squared help?

Answer

Multiple regression predicts one response using two or more predictors at the same time, so it can separate the effect of each input while holding the others steady. R squared always rises when a predictor is added, even a useless one, but adjusted R squared penalizes extra predictors and can fall. A model with five predictors, an R squared of 0.80, and an adjusted R squared of 0.62 is probably carrying predictors that do not earn their place.

#284 · Analyze

When are factor analysis, discriminant analysis, and MANOVA used?

Answer

Factor analysis reduces a large set of correlated variables to a smaller set of underlying factors, which helps a team see the structure behind survey or process data. Discriminant analysis predicts which group a unit belongs to, such as which of three complaint categories a case will fall into. Multivariate analysis of variance tests group differences on several response variables at once, which handles the error rate better than running many separate tests.

#285 · Analyze

What are type I error, type II error, power, and the difference between statistical and practical significance?

Answer

A type I error, called alpha, is rejecting a true null hypothesis and is often set at 0.05, while a type II error, called beta, is failing to reject a false null hypothesis and power is 1 minus beta. Statistical significance means a result is unlikely to be explained by chance alone, and practical significance means the effect is large enough to matter to the business. A large sample can make a 0.01 percent difference significant while the change is not worth the tooling cost.

#286 · Analyze

How is sample size chosen for a test of a mean or of a proportion?

Answer

For a mean, the sample size is driven by the standard deviation, the difference the team wants to detect, and the chosen alpha and power. For a proportion, the sample size equals $z^2 \times p \times (1 - p)$ divided by the error squared. With a 95 percent confidence level, an expected proportion of 0.5, and a margin of error of 0.05, the required sample is 385.

#287 · Analyze

How do confidence, prediction, and tolerance intervals differ, and what are bias and efficiency?

Answer

A confidence interval estimates a range for a population value such as the mean, a prediction interval estimates a range for a single future observation, and a tolerance interval covers a stated share of the population with a stated confidence. The prediction interval is wider than the confidence interval because it adds the scatter of one future point to the uncertainty in the line, while a tolerance interval that covers 95 percent of the parts with 95 percent confidence is usually the widest of the three. An estimator is unbiased when its average over many samples equals the true value, and it is efficient when it has smaller variance than other estimators.

#288 · Analyze

Which hypothesis tests cover means, variances, and proportions?

Answer

The one-sample t test compares a sample mean with a target, the two-sample t test compares two group means, and the paired t test compares before and after readings on the same units. The F test compares two variances, and a z test or proportion test compares a sample proportion with a standard or with another proportion. A team comparing two suppliers of the same part would use a two-sample t test on the critical measurement.

#289 · Analyze

What does analysis of variance do?

Answer

Analysis of variance, or ANOVA, tests whether the means of three or more groups are equal by comparing the variation between group means with the variation within groups. The test statistic F is the mean square between groups divided by the mean square within groups, and a large F with a small p value means at least one group differs. With a between-groups mean square of 30 and a within-groups mean square of 10, F is 3.0, and the team then runs a follow-up test to find which group is different.

#290 · Analyze

How is a chi square test used?

Answer

A chi square goodness of fit test compares observed counts with expected counts to see whether a distribution matches a claim, and a chi square test of independence checks whether two categorical variables are related. The statistic sums the squared difference between observed and expected, divided by expected, across all cells. With observed counts of 30, 20, and 10 against an expected 20 in each of three categories, chi square is 10 with 2 degrees of freedom.

#291 · Analyze

Which types of risk should a Black Belt identify and prioritize?

Answer

The BoK names enterprise risk, operational risk, supplier risk, security risk, product risk, and cybersecurity risk, and a project can also carry schedule, resource, and quality risk. Risks are assessed by how likely they are and how much harm they would cause, then prioritized so the biggest exposures get attention first. A supplier risk might be a single source part, and its countermeasure could be qualifying a second supplier.

#292 · Analyze

How does an FMEA work and what is the risk priority number?

Answer

A failure mode and effects analysis lists each way a process or product can fail, the effect of the failure, and the cause, then scores severity, occurrence, and detection on 1 to 10 scales. The risk priority number is severity times occurrence times detection, so scores of 8, 4, and 3 give a risk priority number of 96. A design FMEA looks at the product design and a process FMEA looks at how the process is run, and the team acts on the highest scores first.

#293 · Analyze

What is gap analysis?

Answer

Gap analysis compares the current state with a desired future state using defined measures, and the gap is the difference between them. It shows how far the process must travel and where the largest shortfalls sit. A service desk with a 70 percent first-contact resolution rate against a target of 90 percent has a 20 point gap to close.

#294 · Analyze

Which root cause analysis tools does a Black Belt use?

Answer

The five whys asks why repeatedly until the chain of causes reaches a root, a cause and effect diagram sorts possible causes into categories such as man, machine, material, method, measurement, and environment, and a fault tree works down from a failure event through logic gates. A Pareto chart ranks causes so the team works on the few that drive most of the problem, and an A3 tells the whole story on one page. A Pareto chart showing that two of six defect types make up 85 of 100 defects tells the team where to start.

#295 · Analyze

What are the seven classic wastes?

Answer

The seven classic wastes are overproduction, inventory, defects, over-processing, waiting, motion, and transportation, and the BoK adds resource under-utilization as an eighth. Overproduction means making more or earlier than the customer needs, and it hides the other wastes behind extra inventory. A work cell that walks 200 steps to fetch a tool every hour is carrying a motion waste that a simple relocation can remove.

#296 · Analyze

What is the difference between an observational study and an experimental study?

Answer

An observational study watches what happens without controlling the variables, while an experimental study deliberately changes one or more factors and holds the rest as steady as possible. Only the experiment supports a strong cause-and-effect claim, because the observational study may be carrying unknown differences between the groups.

#297 · Analyze

What is regression to the mean, and why does it fool improvement teams?

Answer

Extreme readings tend to be followed by less extreme ones even when nothing changes, because part of the extreme value was luck. A team that starts an improvement right after the worst month will usually see a better next month for that reason alone, so it needs a proper baseline instead of a comparison to the worst point.

#298 · Analyze

Why is extrapolating a regression line beyond the data range risky?

Answer

The line is fitted to the range that was actually observed, and the relationship may change outside it. A model built from temperatures between 300 and 400 degrees can predict nonsense at 700 degrees, where the chemistry or the equipment behaves differently.

#299 · Analyze

What is Simpson's paradox, and how can it mislead a team?

Answer

Simpson's paradox appears when a trend in the whole data set reverses once the data are split into groups. A plant can show a higher overall defect rate after an improvement because the mix of products shifted toward harder work, even though every product line improved.

#300 · Analyze

How does measurement error weaken a correlation?

Answer

Random measurement error in either variable adds scatter and pulls the observed correlation toward zero, so the true relationship can be stronger than the reported one. A weak correlation between two noisy lab measurements may understate the real link, which is one reason to study the measurement system first.

#301 · Analyze

What does a control group add to a study?

Answer

A control group experiences everything except the treatment, so the difference between the groups isolates the treatment effect from other changes happening at the same time. Without a control group, a seasonal shift or a general improvement can be credited to the change under study.

#302 · Analyze

How does a two-proportion z test compare two defect rates?

Answer

The test compares the difference between two proportions with the standard error of that difference, then converts the result to a z score and a p value. If one line ships 30 defects in 500 units and another ships 15 in 500, the pooled defect rate is 45 in 1,000, and the test shows whether the gap is larger than chance.

#303 · Analyze

How is a confidence interval for a proportion built?

Answer

The interval centers on the sample proportion and adds and subtracts a margin based on the standard error and the chosen confidence level. With 40 defects in 200 units, the proportion is 0.20, the standard error is about 0.028, and the 95 percent interval runs from roughly 0.14 to 0.26.

#304 · Analyze

What is the margin of error in a survey estimate?

Answer

The margin of error is the plus or minus amount added to a sample estimate to build a confidence interval, driven by the confidence level, the sample size, and the variation. Doubling precision requires roughly four times the sample, because the margin shrinks with the square root of the sample size.

#305 · Analyze

How does an F test compare the variances of two processes?

Answer

The F test divides the larger sample variance by the smaller one and compares the result with an F distribution. A ratio near 1 suggests the variances are similar, while a large ratio suggests one process is far less consistent, which matters before choosing a pooled test.

#306 · Analyze

What is a risk matrix, and how does it prioritize risks?

Answer

A risk matrix plots each risk on two axes, usually likelihood and impact, and sorts the risks into priority bands. Two risks with the same impact but very different likelihood land in different bands, so the team can see which one deserves the first countermeasure.

#307 · Analyze

What is an effect size, and how does it support a practical significance judgment?

Answer

An effect size expresses the size of a difference in standard units, such as the difference between two means divided by the standard deviation. A small p value with a tiny effect size says the difference is real but not large enough to change a decision.

#308 · Analyze

What is a lurking variable, and how does it differ from a confounding variable?

Answer

A lurking variable is one that was not measured but still influences the response, while a confounding variable is tied to the factor being studied so its effect cannot be separated from the factor's effect. A study of shift performance that ignores machine age has a lurking variable, and if older machines run only on the night shift, machine age also confounds the shift comparison.

#309 · Analyze

Why can a correlation of zero still hide a strong relationship?

Answer

The correlation coefficient measures only a straight-line relationship, so a perfect curve can produce a correlation near zero. A U-shaped relationship between temperature and yield would show almost no linear correlation while the effect of temperature is strong and important.

#310 · Analyze

What is the family-wise error rate, and why does it grow with the number of comparisons?

Answer

The family-wise error rate is the chance of at least one false alarm across a set of tests, and it grows as tests are added. Running ten independent tests at a 0.05 level gives a family-wise error rate near 40 percent, which is why ANOVA or an adjusted alpha is used instead of many separate tests.

#311 · Analyze

What does the correlation coefficient tell you?

Answer

The correlation coefficient r ranges from minus 1 to plus 1 and measures the strength and direction of a linear relationship between two variables. A value near plus 1 means a strong positive relationship, near minus 1 means a strong negative relationship, and near 0 means little linear relationship. An r of 0.9 has an r squared of 0.81, so about 81 percent of the variation in one variable is explained by the linear relationship with the other.

#312 · Analyze

Why does correlation not prove causation?

Answer

Two variables can move together because a third hidden variable drives both, because the relationship is coincidence, or because the direction of cause runs the other way. Only a designed experiment with controlled factors can establish cause with confidence. Ice cream sales and drowning rates rise together, but hot weather drives both.

#313 · Analyze

What is a simple linear regression model?

Answer

A simple linear regression predicts a response y from one predictor x using the equation y equals b_0 plus $b_1 x$ plus error. The slope b_1 is the change in y for a one-unit change in x , and least squares chooses the line that minimizes the sum of squared residuals. For y equals 6 plus $2.5x$, a value of x equal to 20 gives a predicted y of 56 .

#314 · Analyze

What do R squared and residual analysis tell you about a regression?

Answer

R squared is the share of variation in the response explained by the model, and adjusted R squared lowers the score when extra predictors do not earn their place. Residual analysis checks whether the leftover errors are random, evenly spread, and roughly normal with no strong outliers. A curved pattern in the residuals means a straight line is the wrong model even when R squared looks high.

#315 · Analyze

What is the difference between a confidence interval and a prediction interval?

Answer

A confidence interval estimates a range for the mean response at a given x , while a prediction interval estimates a range for one future individual value at that x . The prediction interval is always wider because it includes both the uncertainty in the mean and the natural scatter of individual values. A tolerance interval is different again because it covers a stated proportion of the population with a stated confidence.

#316 · Analyze

What are the null hypothesis, the alternative hypothesis, and a type I error?

Answer

The null hypothesis states that there is no effect or no difference, and the alternative states what the analyst hopes to detect. The significance level α is the chance of a type I error, which means rejecting a true null hypothesis. With α equal to 0.05, the test allows a 5 percent chance of declaring an effect that is not real.

#317 · Analyze

What are a type II error, power, and statistical versus practical significance?

Answer

A type II error is failing to reject a false null hypothesis, its probability is beta, and power equals 1 minus beta, the chance of detecting a real effect. Statistical significance means the result is unlikely under the null, while practical significance means the size of the effect matters in the real process. A tiny improvement can be statistically significant with a huge sample yet be too small to justify the cost.

#318 · Analyze

What drives sample size for tests of means and proportions?

Answer

Sample size depends on the size of the effect to detect, the variability, the chosen alpha, and the desired power. Smaller effects, noisier data, or higher power all require larger samples. A team planning a proportion test also needs a baseline rate and a target rate to estimate the needed count.

#319 · Analyze

When do you use a t test, a z test, or ANOVA?

Answer

A t test compares means when the standard deviation is estimated from the sample, and a z test fits when the standard deviation is known or the sample is large. ANOVA compares the means of three or more groups at once. The F statistic in ANOVA is the between-group variation divided by the within-group variation, and a large F suggests at least one group mean differs.

#320 · Analyze

What are chi square tests used for?

Answer

A chi square goodness of fit test checks whether observed counts match an expected distribution, and a chi square test of independence checks whether two categorical variables are related. Both compare observed counts with the counts expected if the null hypothesis were true. A team can test whether defect type and shift are independent by comparing the counts in a table.

#321 · Analyze

What types of risk does a Black Belt assess?

Answer

Risk types include enterprise, operational, supplier, security, product, and cyber security risk. Each one is assessed by how likely it is and how much damage it would cause, then ranked so the largest exposures get attention first. A single supplier for a critical part creates supplier risk, while a stale software patch creates cyber security risk.

#322 · Analyze

What is FMEA and how is the risk priority number calculated?

Answer

Failure mode and effects analysis lists each way a product or process can fail, the effect of that failure, and its likely cause. The team scores severity, occurrence, and detection from 1 to 10, then multiplies them to get the risk priority number. A severity of 7, occurrence of 4, and detection of 3 gives a risk priority number of 84, and the common cutoff of 100 is a local convention rather than an ASQ rule.

#323 · Analyze

How does a design FMEA differ from a process FMEA?

Answer

A design FMEA studies the product design before it is built and looks at failure modes caused by design choices. A process FMEA studies the manufacturing or service process and looks at failure modes caused by process steps, tooling, or methods. Both produce actions, but the design version changes drawings and specifications while the process version changes work instructions and controls.

#324 · Analyze

What root cause tools does a Black Belt use?

Answer

The five whys asks why repeatedly until the chain reaches a system cause rather than a person. A cause and effect diagram, also called a fishbone, sorts possible causes into branches such as man, machine, method, material, measurement, and environment. A fault tree starts from a failure and works backward through AND and OR gates to find the combinations that could produce it.

#325 · Analyze

What is Pareto analysis, and what is gap analysis?

Answer

Pareto analysis ranks causes by frequency or cost and applies the 80/20 idea that a few causes create most of the problem. A team then works on the vital few instead of spreading effort across every cause. Gap analysis compares current performance with a desired future state using set metrics, so a process at 82 percent on-time delivery against a 95 percent target has a 13-point gap.

#326 · Analyze

What does a p-value mean?

Answer

The p-value is the chance of seeing a result at least as extreme as the observed result if the null hypothesis is true, and it is not the chance that the null hypothesis is true. A p-value of 0.03 says that a result this large would occur 3 times in 100 when there is really no effect.

#327 · Analyze

How does a team verify that a candidate root cause is the real driver?

Answer

The team turns the cause on and off and watches the effect, or links it to the outcome with a designed experiment or a regression, rather than accepting a plausible story. A team that assumes the cause without testing may spend the budget fixing the wrong thing.

#328 · Analyze

How does a team choose between a parametric and a nonparametric test?

Answer

Parametric tests assume a distribution shape, usually normal, and use the actual values, while nonparametric tests rank the data and need fewer assumptions. A small sample with a clear outlier points to a rank-based test, while a large normal sample lets the parametric test use more of the information.

#329 · Analyze

What does a team do when an analysis of variance finds a difference but not which groups differ?

Answer

The F test only says that at least one mean differs, so a follow-up comparison such as the Tukey test finds the specific pairs. The follow-up must control the error rate, because running several unadjusted comparisons brings back the false alarm problem the F test avoided.

#330 · Analyze

When does a test use the t distribution instead of the normal distribution?

Answer

The t distribution applies when the population standard deviation is unknown and the sample is small, because it has wider tails that account for the extra uncertainty. As the sample grows, the t distribution approaches the normal, and with large samples the two give nearly the same result.

#331 · Analyze

What do degrees of freedom mean in a statistical test?

Answer

Degrees of freedom are the number of independent pieces of information left after the sample is used to estimate other quantities. In a one-sample t test with 20 observations, estimating the mean leaves 19 degrees of freedom for the standard deviation.

#332 · Analyze

How does a two-way analysis of variance differ from a one-way analysis of variance?

Answer

A one-way study tests one factor with several levels, while a two-way study tests two factors at once and can also test their interaction. A two-way study of machine and shift can show that the machines differ only on the night shift.

#333 · Analyze

What is an influential point in a regression, and why does it matter?

Answer

An influential point is an observation whose removal changes the fitted model substantially, often because it sits far from the other values in the input direction. A single order with an extreme quantity can define the slope of a regression line for the whole data set.

#334 · Analyze

What is a dummy variable in a regression model?

Answer

A dummy variable codes a category as 0 or 1 so it can enter a numeric model, such as 1 for the new process and 0 for the old one. Its coefficient estimates the average shift in the response when the category is present, holding the other predictors constant.

#335 · Analyze

How does a confidence interval relate to a hypothesis test?

Answer

A two-sided test at a significance level matches a confidence interval at the same level, so if the interval excludes the null value, the test rejects the null. A 95 percent interval for the difference from 2.1 to 4.6 excludes zero and matches a significant two-tailed test at 0.05.

#336 · Analyze

What rule of thumb applies to expected counts in a chi square test?

Answer

The test works best when each expected count is at least 5, because small expected counts make the statistic unstable and the p-value unreliable. A table with an expected count of 1.2 in one cell may need more data or a different test.

#337 · Analyze

How does a cause differ from a symptom in root cause analysis?

Answer

A symptom is the visible effect, such as late deliveries, and a cause is the condition that produces it, such as a release rule that starts work too late. Fixing the symptom removes the signal for a while, and the problem returns.

#338 · Analyze

When does a team stop digging for a root cause?

Answer

The team stops when the cause is specific enough to act on and the evidence supports it, not when the answers run out. A five whys chain that ends at the operator and never tests the claim has stopped too early.

#339 · Analyze

Why does a high correlation not prove that two measurement methods agree?

Answer

Correlation measures whether two sets of values move together, while agreement measures whether they give the same value, and two methods can correlate strongly and still differ by a constant bias. Two gauges that differ by exactly 2 units on every part have a correlation of 1.0 and poor agreement.

#340 · Analyze

What weakness does a simple before-and-after comparison share?

Answer

A before-and-after comparison shows the change between two periods, and its weakness is that any other change in the same period could explain the difference. Adding a control site or a second measure separates the project effect from the background drift.

#341 · Analyze

How is a two sample t test calculated and interpreted?

Answer

The test compares two sample means against the standard error of their difference, which depends on the variation within each group and the size of each group. Two lines with means of 12.5 and 10.5, a standard deviation of 2, and 16 samples each give a t value of 2 divided by 0.707, or 2.83, with 30 degrees of freedom. The team compares that value with the critical value for the chosen alpha before claiming that the lines differ.

#342 · Analyze

How is a paired t test set up, and what does it control?

Answer

A paired test works with the differences between matched readings, such as before and after measurements on the same unit, and tests whether the average difference is zero. Nine operators timed with the old and new method might show an average time saving of 4.2 minutes with a standard deviation of 3.0, giving a t value of 4.2 divided by 1.0, or 4.2, with 8 degrees of freedom. Pairing removes the variation between operators, so the test isolates the effect of the method.

#343 · Analyze

How is a two proportion test carried out?

Answer

The test combines the two sample proportions into one pooled estimate, then divides the difference between the proportions by the standard error built from that pooled value. Two shifts with 30 of 120 and 18 of 120 defects give a pooled proportion of 0.20 and a standard error of 0.052, so z works out to about 1.94. The team compares that value with the critical value to decide whether the shifts really differ.

#344 · Analyze

What does an F test for equal variances compare?

Answer

The F test divides the larger sample variance by the smaller one and compares the result with an F distribution that has one degree of freedom count for each sample. Variances of 9 and 4 give an F value of 2.25, which the team checks against the critical value for the two sample sizes. The test matters because several standard methods, including the pooled two sample t test, assume that the groups have similar spread.

#345 · Analyze

How is the F ratio in an analysis of variance built?

Answer

The F ratio divides the variation between group means by the variation within the groups, with each part divided by its own degrees of freedom. If the between group mean square is 40 and the within group mean square is 10, F is 4, which suggests that the group means differ by more than chance alone would produce. A large F means the factor moved the response, while an F near 1 means the groups look like random draws from one pool.

#346 · Analyze

How is a chi square goodness of fit statistic computed?

Answer

For each category, the team subtracts the expected count from the observed count, squares the difference, and divides by the expected count, then adds the categories. With observed counts of 30, 25, and 45 against an expected 33.3 in each of three categories, the statistic is about 6.50 with 2 degrees of freedom. That sits above the 5.99 critical value at the 0.05 level, so the observed pattern does not fit the expected one.

#347 · Analyze

How is a 95 percent confidence interval for a mean computed and read?

Answer

The interval is the sample mean plus and minus the standard error multiplied by the t value for the chosen confidence level and degrees of freedom. A mean of 50 with a sample standard deviation of 10 and 25 readings gives a standard error of 2, so the interval runs from about 45.9 to 54.1. The team reads this as the range of plausible values for the true mean, not as the range that contains 95 percent of future readings.

#348 · Analyze

What is a fault tree analysis, and when is it used?

Answer

A fault tree starts with an undesired top event and works downward through the combinations of failures that could produce it, using AND and OR gates. If two independent failures each have a 0.10 chance and both are needed, the AND gate gives 0.10 times 0.10, or 0.01, while an OR gate for either one gives 1 minus 0.90 times 0.90, or 0.19. The method suits high stakes systems such as medical devices or aircraft, where a team must show which failure paths matter most.

#349 · Analyze

What does an A3 report contain?

Answer

An A3 report fits a problem story on one large sheet, moving from background and current state to the analysis, the countermeasures, and the follow up plan. The left side usually carries the problem, the facts, and the cause, while the right side carries the actions, the owner, and the date. Its purpose is to make the reasoning visible so a reviewer can challenge the logic before money is spent.

#350 · Analyze

How is a cause and effect diagram built with a team?

Answer

The team writes the effect in a box on the right, draws a spine to it, and adds branches for categories such as people, machine, method, material, measurement, and environment. Members place possible causes on the branches and then use data to test which ones actually move the effect. The diagram is a way to organize a brainstorm, so a finished diagram is a list of suspects rather than a proven cause.

#351 · Analyze

What do the severity, occurrence, and detection scales each rate in an FMEA?

Answer

Severity rates how much harm the failure causes if it reaches the customer, occurrence rates how often the cause is expected to happen, and detection rates how well the current controls catch the failure before it escapes. Each scale usually runs from 1 to 10, and a rating of 10 for detection means the failure is nearly certain to escape. A design with severity 8, occurrence 6, and detection 4 has a risk priority number of 192.

#352 · Analyze

What belongs in a risk register, and how is it used?

Answer

A risk register lists each risk, its cause, its potential effect, the likelihood and impact ratings, the owner, and the planned response. A project risk such as a single source sensor supplier might carry a high impact rating and a response of qualifying a second supplier before the pilot. The register is reviewed at each toll gate so risks that grew or faded are re ranked.

#353 · Analyze

How does risk mitigation differ from contingency planning?

Answer

Mitigation reduces the chance or the impact of a risk before it happens, while contingency planning decides in advance what the team will do if the risk still occurs. A team mitigates a supplier risk by qualifying a second source, and it plans a contingency of holding two weeks of safety stock in case the first source still fails. Mitigation changes the risk itself, and contingency changes the response.

#354 · Analyze

What is the standard error, and how does it differ from the standard deviation?

Answer

The standard deviation describes how spread out individual readings are, while the standard error describes how much a sample statistic such as the mean would vary from sample to sample. A process with a standard deviation of 12 and 36 readings has a standard error of 12 divided by 6, or 2. The standard error is the smaller number in large samples, which is why a mean from a big sample is a more precise estimate.

#355 · Analyze

How does a one tailed test differ from a two tailed test?

Answer

A one tailed test places the whole rejection area in one direction, while a two tailed test splits it across both directions and asks only whether a difference exists. The same statistic that gives a one tailed probability of 0.03 gives a two tailed probability of 0.06, so the two tailed test needs a stronger result to reject the null hypothesis. The team should choose the tails before seeing the data, based on whether only one direction matters.

#356 · Improve

What are the basic terms of design of experiments?

Answer

A factor is an input being tested, a level is a setting of that factor, and a response is the measured output. A treatment is one combination of factor levels given to a test unit, error is the variation not explained by the factors, and nested means one factor exists only within the levels of another factor. The independent variable is the factor the team sets and the dependent variable is the response it measures.

#357 · Improve

Which principles shape the design of an experiment?

Answer

Randomization assigns the runs in random order to protect against unknown sources of drift, replication repeats the whole run to estimate error, repetition takes more than one reading from the same run, and blocking groups similar units to remove a known nuisance factor. Balance keeps the levels even, confounding mixes one effect with another, and resolution describes how badly effects are confounded. Power and sample size set the chance of detecting a real effect.

#358 · Improve

How is an experiment planned?

Answer

Planning starts with a clear objective, then selects the factors, the levels, the response, and the measurement method, and finally chooses the design. The team also decides the run order, the number of replicates, and how the data will be analyzed before any data is collected. An experiment planned without a defined response often ends with numbers that cannot answer the question.

#359 · Improve

When are completely randomized, randomized block, and Latin square designs used?

Answer

A completely randomized design assigns treatments to units at random and is used when the units are alike. A randomized block design first groups similar units into blocks, such as shifts or lots, then randomizes the treatments within each block to remove the block effect. A Latin square design controls two nuisance factors at once by arranging treatments in a square so each treatment appears once in every row and every column.

#360 · Improve

What is a full factorial experiment?

Answer

A full factorial experiment runs every combination of the chosen factor levels, so it can estimate every main effect and every interaction. The number of runs is the product of the levels, so three factors at two levels each need 2 times 2 times 2, or 8 runs. A welding study with two temperatures and two pressures runs four combinations and can show whether the two factors interact.

#361 · Improve

What is a two-level fractional factorial and what is confounding?

Answer

A fractional factorial runs only a chosen fraction of the full set, which saves time and cost when many factors must be screened. The trade is confounding, where one effect is mixed with another effect so the two cannot be separated from the data, and resolution describes how severe that mixing is. Five factors at two levels would need 32 runs in a full factorial but can be screened in 16 runs with a resolution that keeps the main effects clear of two-factor interactions.

#362 · Improve

How do pull systems and kanban control production?

Answer

A pull system starts work only when a downstream customer signals a need, so material does not build up between steps. Kanban is the signal itself, often a card or an empty container, and the number of cards caps the work in process. With a daily demand of 200 units, a two day replenishment lead time, a 1.2 safety factor, and a container of 50 units, the card count is $200 \times 2 \times 1.2$ divided by 50, which is 9.6 and rounds up to 10 cards.

#363 · Improve

What is 5S?

Answer

5S is a workplace organization method whose five steps are sort, set in order, shine, standardize, and sustain. The steps remove what is not needed, give everything a place, clean the area, set the standard, and keep the practice going. A tool crib that labels every drawer and marks the floor for carts cuts search time and makes a missing tool obvious at a glance.

#364 · Improve

What are standard work and poka-yoke?

Answer

Standard work documents the current best method: the takt time, the work sequence, and the standard work in process, so every operator performs the task the same way. Poka-yoke is a mistake-proofing device or check that prevents an error or makes it obvious right away. A fixture that only lets a part load one way prevents a reversed assembly before it can happen.

#365 · Improve

Which tools reduce cycle time?

Answer

Continuous flow moves a unit through the process with as little waiting as possible, single-minute exchange of die cuts changeover time by separating internal work that stops the machine from external work that can be done while it runs, and heijunka levels the mix and volume of production so the process does not swing between feast and famine. A changeover that takes 90 minutes and moves 60 minutes of work to external preparation leaves only 30 minutes of downtime, a 67 percent reduction. Shorter changeovers allow smaller batches and faster response to customer demand.

#366 · Improve

How does kaizen differ from a kaizen blitz?

Answer

Kaizen is continuous, incremental improvement carried out by the people who do the work, while a kaizen blitz is a short, focused event, often a few days, that attacks one defined problem with a dedicated team. Kaizen builds a habit over time and a blitz produces a fast, visible change. A plant might hold a blitz to reorganize a work cell in three days, then use kaizen to keep refining it every week.

#367 · Improve

What is the theory of constraints?

Answer

The theory of constraints says that a system output is limited by one bottleneck at a time, so improvement effort should go to that constraint first. The steps are to identify the constraint, exploit it, subordinate everything else to it, elevate it, and then repeat once it moves. A machine that takes 25 minutes per unit and runs 420 minutes a day can produce only 16.8 units a day, so faster work elsewhere just builds inventory in front of it.

#368 · Improve

What does overall equipment effectiveness measure?

Answer

Overall equipment effectiveness multiplies availability, performance, and quality to show how much of the planned production time produced good parts at full speed. Availability is running time divided by planned time, performance compares actual speed with ideal speed, and quality is good units divided by total units. A line at 90 percent availability, 95 percent performance, and 98 percent quality has an overall equipment effectiveness of 0.90 times 0.95 times 0.98, or about 83.8 percent.

#369 · Improve

How should an improvement be implemented?

Answer

Implementation plans describe the change, the owner, the timeline, the training, and the measures, and they are tested first with a pilot or a simulation before full rollout. The pilot is run on a limited area or shift so the team can compare results with the current process and catch problems cheaply. A hospital that pilots a new discharge checklist on one ward for two weeks can fix the wording before spreading it to every ward.

#370 · Improve

What does a nested factor mean in a designed experiment?

Answer

A nested factor appears only inside one level of another factor, so its levels are not the same across the whole experiment. If three operators run batches and each operator uses a different machine, machine is nested within operator, and the analysis must be set up that way or the error term will be wrong.

#371 · Improve

How is a kanban card count calculated from demand, lead time, and container size?

Answer

The count equals average demand multiplied by replenishment lead time, times a safety factor, divided by the container size. With demand of 300 units a day, a lead time of 4 days, and containers of 50 units, the base count is 300 times 4 divided by 50, which is 24 cards.

#372 · Improve

What does an interaction between two factors look like?

Answer

An interaction means the effect of one factor depends on the level of the other factor. If a catalyst speeds up the reaction at high temperature but slows it at low temperature, temperature and catalyst interact, and reporting two separate main effects would hide the real behavior.

#373 · Improve

How does a team choose the response to measure in an experiment?

Answer

The response is the measure used to judge the treatment, so it must connect to the customer requirement, respond quickly, and be measurable with acceptable error. Measuring only yield when the customer cares about strength can produce a design that meets the yield target and still fails in service.

#374 · Improve

What is the difference between replication and repetition?

Answer

Replication repeats an entire treatment as a separate randomized run, which gives an independent estimate of the error. Repetition takes several readings from the same run, which measures the measurement system but cannot show the run-to-run variation.

#375 · Improve

How does a team decide how many replicates an experiment needs?

Answer

The number of replicates follows from the size of the effect the team needs to detect, the amount of noise in the response, and the accepted risk of a wrong conclusion. A larger effect or a quieter process needs fewer replicates, and power software or a standard formula converts those inputs into a run count.

#376 · Improve

How does blocking differ from nesting in a designed experiment?

Answer

Blocking deliberately groups runs that share a known nuisance factor, such as the day or the batch of raw material, so that factor is removed from the error estimate. Nesting happens when the levels of one factor exist only inside one level of another, such as machines used by only one operator, and it changes how the error term is formed.

#377 · Improve

How does a team choose between two improvement options that both work?

Answer

The team scores both options on the same criteria, such as cost, cycle time, quality, risk, and ease of sustainment, and weights those criteria before comparing totals. Running a small pilot of each option gives real evidence instead of a debate based on opinions.

#378 · Improve

How is the payback period of an improvement calculated?

Answer

The payback period divides the one-time cost of the improvement by the savings per period. A 12,000 dollar fixture that saves 3,000 dollars a month pays back in 4 months, which is easy for a sponsor to compare with the payback of another option.

#379 · Improve

How is the value-added ratio of a process calculated?

Answer

The value-added ratio divides the time that actually changes the product or delivers the service by the total lead time. A process with 12 minutes of value-added work inside a 60-minute lead time has a ratio of 20 percent, which shows how much of the customer's wait is spent moving, queuing, or waiting for approval.

#380 · Improve

What is the difference between a push system and a pull system?

Answer

A push system builds to a forecast and moves work downstream whether or not the next step is ready, while a pull system releases work only when the downstream step signals that it can accept more. Pull systems limit work in progress, which shortens the queue and exposes problems instead of burying them in inventory.

#381 · Improve

How does continuous flow reduce cycle time compared with batch processing?

Answer

Continuous flow moves one unit through the steps with as little waiting as possible, while batch processing makes units wait until a whole lot is ready for the next step. Cutting the batch size from 500 to 50 shortens the wait at each step, so the same work finishes in far less lead time.

#382 · Improve

How does a simulation support an improvement decision before any equipment changes?

Answer

A simulation builds a model of the process, runs different demand and staffing scenarios, and reports the results without touching the real line. A team can test whether an extra nurse at 7 a.m. shortens the emergency wait before committing to the schedule change.

#383 · Improve

How does a team decide when an improvement is ready for full rollout?

Answer

An improvement is ready when the pilot met the target, the control plan is in place, the operators are trained, and the process owner accepts responsibility. Rolling out on a promising pilot result alone moves an unproven risk from a small area to the whole plant.

#384 · Improve

What do the basic DOE terms mean?

Answer

A factor is an input variable the team changes, a level is a setting of that factor, and a response is the output being measured. A treatment is one combination of factor levels, replication repeats a whole run to estimate error, and repetition takes repeated readings within the same run. Error is the variation not explained by the factors.

#385 · Improve

What are the main DOE design principles?

Answer

Randomization spreads the effect of nuisance factors so they do not line up with a factor of interest. Blocking groups runs that share a known nuisance condition, replication gives an independent estimate of error, and balance keeps the design from favoring one factor level. Sample size and power describe how much data the experiment needs to detect a real effect.

#386 · Improve

What is confounding, and what does resolution mean in a fractional design?

Answer

Confounding happens when the effect of one factor cannot be separated from the effect of another factor or an interaction. Resolution III designs keep main effects clear of each other but confound them with two-factor interactions, resolution IV designs keep main effects clear of two-factor interactions but confound some two-factor interactions with each other, and resolution V designs keep main effects and two-factor interactions clear of each other. A higher resolution design needs more runs.

#387 · Improve

How does a team plan a designed experiment?

Answer

The team states the objective, selects the factors and levels, defines the response and how to measure it, chooses a design, sets sample size and power, and decides the run order. Poor planning produces data that cannot answer the question, so the team also checks the measurement system before running. A pilot run can catch setup problems before the full experiment.

#388 · Improve

How does a full factorial design differ from a fractional factorial design?

Answer

A full factorial runs every combination of factor levels, so three factors at two levels need 8 runs and can estimate all main effects and interactions. A fractional factorial runs only a chosen subset, so a half fraction of three factors needs 4 runs and screens many factors with less data. The tradeoff is that some effects become confounded.

#389 · Improve

When do you use completely randomized, randomized block, and Latin square designs?

Answer

A completely randomized design assigns treatments randomly to units when the units are fairly uniform. A randomized block design first divides units into similar blocks, then randomizes treatments inside each block when one nuisance variable is known. A Latin square controls two nuisance variables at once and uses one treatment factor.

#390 · Improve

What is response surface methodology?

Answer

Response surface methodology models curvature in the response and finds factor settings that optimize it. Designs such as the central composite design combine factorial points, axial points, and center points so the team can fit a curved model. It usually follows a screening experiment that already narrowed the important factors.

#391 · Improve

What is a work cell, and how does it differ from a functional layout?

Answer

A work cell groups the people and machines needed to make a family of similar products in one place and in sequence, so parts travel short distances and wait less. A functional layout groups like machines together, such as every lathe in one department, which forces parts to move between departments and queue at each stop. Rearranging a job shop into two cells can cut a part's travel from 800 feet to 60 feet and remove most of the waiting.

#392 · Improve

How do pull systems and kanban work?

Answer

A pull system starts work only when downstream demand signals that more is needed, which limits overproduction and work in progress. Kanban cards or signals are the trigger that authorizes replenishment. With demand of 300 units per day, a 2-day lead time, 100 units of safety stock, and a container size of 50, the kanban count is $600 \text{ plus } 100, \text{ divided by } 50, \text{ or } 14 \text{ cards}$.

#393 · Improve

What do muda, mura, and muri mean in lean thinking?

Answer

Muda is waste, the work itself that adds no value to the customer, such as waiting, rework, or extra motion. Mura is unevenness, such as a schedule that swings between idle time and overload, and muri is overburden, asking a person or machine to work beyond a safe or sustainable pace. A line that builds 200 units one day and 900 the next has mura even when no single step looks wasteful.

#394 · Improve

How do internal and external setup differ in a quick changeover?

Answer

Internal setup work can only be done while the machine is stopped, such as removing a die or aligning a new fixture. External setup work can be done while the machine is still running, such as fetching tools, paperwork, or prepared materials, and moving those tasks outside the stop cuts changeover time with no capital cost. A press changeover that drops from 60 minutes to 18 minutes often moves most of the searching and staging into the external category.

#395 · Improve

What are SMED, heijunka, and continuous flow?

Answer

Single-minute exchange of die cuts changeover time by moving work from internal setup, which stops the machine, to external setup, which happens while the machine runs. Heijunka levels production by volume and mix so the process avoids peaks and valleys. Continuous flow arranges steps so work moves one piece at a time without waiting.

#396 · Improve

What is a confirmation run after a designed experiment?

Answer

A confirmation run repeats the predicted best settings on a fresh set of parts to check that the model's prediction holds outside the original runs. Agreement between the predicted and observed result supports a wider rollout, while a large gap warns that the model missed a factor or that the process drifted. If the model predicts a yield of 92 percent and the confirmation run returns 84 percent, the team should study the gap before changing the process for good.

#397 · Improve

What are the theory of constraints, overall equipment effectiveness, and pilot testing?

Answer

The theory of constraints improves the system by finding the bottleneck, exploiting it, subordinating other steps to it, elevating its capacity, and then repeating the cycle. Overall equipment effectiveness multiplies availability, performance, and quality, so 0.90 times 0.95 times 0.98 gives 83.8 percent. Before a full rollout, a team runs a pilot test or simulation to confirm the change works and to catch problems on a small scale.

#398 · Improve

How does a team choose the levels of a factor in an experiment?

Answer

The levels must sit far enough apart to show an effect but stay inside the safe and usable operating range, and one level is usually the current setting. Testing 300 and 350 degrees on an oven that can hold only 320 wastes runs and teaches nothing about the high setting.

#399 · Improve

A line is planned for 480 minutes, is down 48 minutes, makes 900 units against an ideal of 960, and 855 are good. What is OEE?

Answer

Availability is 432 divided by 480, which is 0.90, performance is 900 divided by 960, which is 0.9375, and quality is 855 divided by 900, which is 0.95. Multiplying the three gives 0.8016, so the overall equipment effectiveness is about 80.2 percent.

#400 · Improve

What is a supermarket in a pull system?

Answer

A supermarket is a small controlled store of parts that downstream cells withdraw from, and the withdrawal signals upstream to replenish what was taken. The cell keeps a small market of common parts instead of a warehouse, and the market size sets the cap on inventory.

#401 · Improve

How does line balancing relate to takt time?

Answer

Line balancing assigns work elements to stations so each station's cycle time fits under takt time and the work is shared evenly. If takt is 36 seconds and one station needs 40 seconds, that station is the bottleneck and the line misses demand even when the other stations have idle time.

#402 · Improve

How does a drum buffer rope system schedule a constraint?

Answer

The drum sets the pace of the whole system, the buffer protects the constraint with time or material so it never starves, and the rope releases new work at the drum's pace. Work released faster than the drum only builds inventory in front of the constraint.

#403 · Improve

What is yokoten?

Answer

Yokoten is the horizontal spread of a proven improvement to other lines, plants, or shifts, with local adaptation instead of a blind copy. A fix proven on line 1 is documented and copied to lines 2 and 3 after each team checks its own conditions.

#404 · Improve

What is a nuisance variable, and how does it differ from a factor?

Answer

A nuisance variable affects the response but is not part of the question, such as the humidity in the room or the operator on the shift. The team blocks or randomizes against it instead of studying it, while a factor is deliberately varied to learn its effect.

#405 · Improve

When does an experiment need a split-plot design?

Answer

A split-plot design suits a study where one factor is hard to change and the other is easy, so the hard factor is set for a group of runs and the easy factor is varied inside each group. A furnace study that must hold the furnace at each temperature for a whole day uses a split-plot layout.

#406 · Improve

What is a covariate, and how does it help an experiment?

Answer

A covariate is a measured variable that is not the factor of interest but explains part of the variation in the response, such as the incoming moisture of raw material. Measuring it and adding it to the model reduces the error term, so a smaller effect becomes visible.

#407 · Improve

Why must a team separate the experimental unit from the measurement unit?

Answer

The experimental unit is the thing that receives a treatment, and the measurement unit is the thing that is measured, so counting several readings from one unit as independent inflates the sample size. Treating ten readings from one batch as ten independent runs hides the batch-to-batch variation.

#408 · Improve

How does a fixed factor differ from a random factor?

Answer

A fixed factor has specific levels the team wants to compare, such as three named machines, while a random factor is a sample from a larger population of levels, such as three machines chosen from twenty. The distinction changes which conclusions the study can support.

#409 · Improve

When does a team use a D-optimal design?

Answer

A D-optimal design is built by computer to fit awkward constraints, such as factors that cannot be set independently or a limited number of runs. A study that can only run certain combinations uses D-optimal methods instead of a standard design.

#410 · Improve

What is jidoka?

Answer

Jidoka means building quality into the process by giving the machine or the operator the ability to stop work when something abnormal appears, so the problem is fixed before bad units pile up. An andon cord that halts the line is jidoka in action.

#411 · Improve

What is an andon?

Answer

An andon is a visible signal, often a light or a board, that calls for help or shows the state of a station. A green light for normal and a red light for a stop let a supervisor see the state of the whole line from one place.

#412 · Improve

How is a main effect calculated from a two level experiment?

Answer

The main effect is the average response at the high level minus the average response at the low level, averaged across every run at those levels. If temperature runs average 50 at the low setting and 56 at the high setting, the temperature main effect is plus 6 units. A main effect near zero means the factor had little influence on the response over the range tested.

#413 · Improve

What is an interaction between two factors, and how is it spotted?

Answer

An interaction exists when the effect of one factor depends on the level of another factor. If raising temperature adds 8 units of response at low pressure but only 1 unit at high pressure, the two factors interact, and the team cannot describe the temperature effect with a single number. Interaction plots show non parallel lines, and the analysis of variance tests whether that pattern is larger than the noise.

#414 · Improve

What is a screening experiment, and what does it accomplish?

Answer

A screening experiment studies many factors at two levels each with a small number of runs, so the team can separate the few factors that matter from the many that do not. Seven factors screened with a 2 to the 7 minus 3 design need only 16 runs instead of 128 for the full factorial. The survivors then move into a fuller experiment where their effects and interactions are estimated with more precision.

#415 · Improve

What is a central composite design, and why does it add axial and center points?

Answer

A central composite design starts with a two level factorial, adds axial points outside the cube to estimate curvature, and repeats center points to estimate pure error. A two factor version uses 4 cube points, 4 axial points, and 5 center points for 13 runs. The axial points let the team fit a curved response surface and locate a setting near the optimum rather than only comparing corners.

#416 · Improve

What are the five focusing steps of the theory of constraints?

Answer

The steps are to identify the constraint, exploit it so it never starves, subordinate everything else to that decision, elevate the constraint with more capacity, and then start again because the constraint will have moved. A plant that finds its constraint at final test runs the upstream steps to a schedule that keeps final test fed and busy. The cycle repeats because solving one constraint usually exposes the next one.

#417 · Improve

What are the six big losses that overall equipment effectiveness exposes?

Answer

The losses group into breakdowns and setup time under availability, idling and minor stops and reduced speed under performance, and process defects and reduced yield under quality. A line that loses two hours to a breakdown and one hour to a changeover has paid for that time in the availability factor. Naming the loss tells the team which improvement tool to use instead of only reporting a low percentage.

#418 · Improve

How does a team build a current state and then a future state value stream map?

Answer

The team walks the process from customer order to delivery, records the process time and the waiting time between steps, and draws the information flow that schedules the work. The current state map makes the waiting visible, so a 60 minute lead time with 12 minutes of work shows a value added ratio of 20 percent. The future state map then removes the largest waiting blocks and sets the target flow that the team will implement.

#419 · Improve

What are the three elements of standard work?

Answer

Standard work defines the takt time the cell must meet, the exact work sequence each operator follows, and the standard amount of inventory needed to keep the work flowing. A cell with a two minute takt, a documented sequence of five steps, and two units of standard work in process gives every operator the same picture of a normal cycle. Standard work is the baseline that a kaizen event improves, so it must be written down and posted.

#420 · Improve

How does heijunka level production, and what problem does it solve?

Answer

Heijunka smooths the mix and volume of work over a period instead of building one product in a large batch and then switching to another. A plant that needs 600 units of A and 600 of B each day might alternate small lots through the day rather than run 1,200 of A on Monday. Leveling protects the upstream steps from a demand spike, which lowers the inventory they would otherwise carry.

#421 · Improve

What must be decided before a pilot test starts?

Answer

The team defines the measure, the success threshold, the duration, the sample size, and the conditions under which the pilot will be judged before the first run. A team testing a new intake script might agree in advance that the pilot passes if the error rate falls below 1.5 percent over four weeks on one team. Deciding the rules first prevents the team from explaining away a weak result afterward.

#422 · Improve

When is a computer simulation a better choice than a pilot?

Answer

Simulation fits when a pilot would be too expensive, too slow, or too risky to run, and when the process can be described well enough to model. A hospital considering a new emergency department layout can simulate patient flow for thousands of arrivals instead of rebuilding a wing. The model is only as good as its assumptions, so the team checks it against current performance before trusting its answer.

#423 · Improve

How does poka yoke differ from 100 percent inspection?

Answer

Poka yoke makes an error impossible or immediately obvious at the point where it happens, while 100 percent inspection still relies on a person or device catching a defect after it is made. A fixture that only accepts the part in the correct orientation prevents the error, whereas inspecting every part still lets a tired operator pass a bad one. The device is preferred because it removes dependence on attention.

#424 · Improve

What does a value added ratio measure, and how is it calculated?

Answer

The value added ratio divides the time spent doing work the customer would pay for by the total lead time through the process. A 60 minute lead time with 12 minutes of value added work gives a ratio of 12 divided by 60, or 20 percent. The low percentage shows how much of the delay is waiting, which is the first target for flow improvement.

#425 · Improve

How does a team decide whether a proven improvement should be rolled out to other sites?

Answer

The team checks whether the other sites have the same root cause, the same process conditions, and the resources the improvement needs. A fix that worked because one plant had a dedicated maintenance technician may fail at a site without that role, so the rollout plan names what must be in place first. Rollout also needs a standard method and a way to measure whether each site actually adopts it.

#426 · Control

What are the objectives of statistical process control?

Answer

Statistical process control monitors and controls process performance, tracks trends and runs, and reduces variation by separating common cause from special cause. It uses control charts built from process data rather than from customer specifications. A stable chart tells the team the process is predictable, so any change in the numbers points to something that happened.

#427 · Control

How are critical process characteristics chosen for control chart monitoring?

Answer

The team picks the characteristics that most affect the customer, the output, or the cost, and that can be measured reliably. Key product features, safety items, and the process settings that drive them are the usual choices. A filling line may chart fill weight because it affects both the label claim and the amount of product given away.

#428 · Control

What is rational subgrouping?

Answer

Rational subgrouping means forming the subgroups so that the variation within a subgroup reflects only common cause and the variation between subgroups can show special cause. Subgroups are usually consecutive units made close together under the same conditions. If a team mixes day shift and night shift units inside one subgroup, the shift difference is hidden and the chart loses its power to detect it.

#429 · Control

How is the right control chart selected?

Answer

For variable data measured on a continuous scale, an $\bar{X}$ -bar and R chart suits subgroups of about two to ten, an $\bar{X}$ -bar and s chart suits larger subgroups, and an individuals and moving range chart suits one measurement at a time. For attribute data, a p chart tracks the proportion defective with a varying subgroup size, an np chart tracks the count defective with a constant size, a c chart tracks defects per unit with a constant area, and a u chart tracks defects per unit with a varying area. Short production runs use short-run SPC, and moving average or exponentially weighted moving average charts smooth the data to catch small shifts.

#430 · Control

How does a control chart show common and special cause?

Answer

A point outside a control limit, or a run that breaks a standard rule such as eight points in a row on one side of the center line, signals special cause variation that should be investigated. Points that stay inside the limits with a random pattern show common cause variation that belongs to the system. Reacting to common cause as if it were special cause is tampering and usually makes the variation worse.

#431 · Control

How do control limits differ from specification limits?

Answer

Control limits come from the process itself, usually the mean plus and minus three standard deviations of the plotted statistic, and they show what the process can do. Specification limits come from the customer or the design and show what the process must do. A process can sit comfortably inside its control limits and still produce out-of-specification parts when the process spread is wider than the specification.

#432 · Control

When are short-run SPC and moving average charts used?

Answer

Short-run SPC standardizes the plotted values so several short jobs with different targets can share one chart, which suits job shops and low volume production. Moving average and exponentially weighted moving average charts combine recent points to smooth noise and detect small shifts faster than a plain individuals chart. A machine shop that runs ten different part numbers each week can use short-run methods instead of a separate chart for every part.

#433 · Control

What is total productive maintenance?

Answer

Total productive maintenance builds equipment reliability through operator ownership of daily care, planned and preventive maintenance, and work that improves equipment rather than only repairing it. Overall equipment effectiveness is its main measure, and the goal is to remove breakdowns and speed losses from the process. An operator who checks oil and air pressure each shift can catch a problem before it stops the line.

#434 · Control

What are visual controls?

Answer

Visual controls make the state of a process obvious at a glance, using pictures of the correct procedure, color-coded parts, indicator lights, marked floor locations, and boards that show schedule and status. They let anyone see a problem without reading a report. An andon light that turns on when an operator needs help brings a response without a phone call.

#435 · Control

What belongs in a control plan, and why is the measurement system reviewed again?

Answer

A control plan names each characteristic to be monitored, the specification, the measurement method, the sample size and frequency, who does the measuring, and the reaction when something goes out of control. It transfers responsibility from the project team to the process owner when the project closes. As the process improves, the measurement system takes a larger share of the remaining variation, so a gauge that was good enough before may need to be rechecked and improved.

#436 · Control

How are improvements sustained after the project ends?

Answer

Teams sustain gains by documenting lessons learned, updating standard operating procedures and work instructions, training process owners and staff on the revised method, and continuing to evaluate the process with control charts and control plans. Ongoing evaluation watches leading indicators that predict trouble and lagging indicators that confirm results. A project that ends without a trained owner and an updated procedure usually drifts back to the old way within months.

#437 · Control

What is the difference between a control plan and a standard operating procedure?

Answer

A standard operating procedure tells an operator how to run a step, while a control plan tells the team what to measure, how often, with what limits, and what to do when the measurement signals a change. The procedure describes the method and the control plan describes the surveillance of that method.

#438 · Control

How does a team handle a critical characteristic that can only be tested destructively?

Answer

When a test destroys the unit, the team cannot measure every part, so it relies on a sampling plan and on process controls that predict the result. It also shifts inspection upstream, checking the inputs and settings that determine the destructive property instead of only the finished part.

#439 · Control

How does a team protect a gain when the champion or Black Belt leaves the company?

Answer

The gain survives a departure when the standard work, the control plan, the training, and the process owner's authority are already in place before the leader goes. A project that depended on one person's attention should be documented and handed over before that person's last day.

#440 · Control

How does a team set the first control limits for a process that has never been charted?

Answer

The team collects a baseline of 20 to 25 subgroups from a period when the process was believed to be stable, then calculates the mean and the average range or standard deviation. Those values set trial limits, and any out-of-control points are investigated before the limits are frozen for routine use.

#441 · Control

Which documents must be updated to hold a gain?

Answer

The team updates the standard operating procedures, the work instructions, the process map, the control plan, and any training material that describes the old method. A gain disappears quickly when the old version of the procedure is still the one posted at the workstation.

#442 · Control

What training does a process owner need before taking over a chart?

Answer

The owner needs to read the chart, apply the reaction plan, tell common cause from special cause, and know whom to call when a signal appears. Training ends with the owner demonstrating the routine on live data, not with a signed attendance sheet.

#443 · Control

What are lessons learned, and how do they get reused?

Answer

Lessons learned record what worked, what failed, and what the next team should do differently, captured while the project is still fresh. They get reused when they are stored where other teams can search them and when a standing review checks that new projects apply them.

#444 · Control

How does ongoing evaluation use leading and lagging indicators after handoff?

Answer

Leading indicators, such as the percent of audits completed or the number of charts reviewed on time, warn that the control system is slipping. Lagging indicators, such as the defect rate or the cost per unit, confirm whether the gain is still real, and together they separate a genuine decline from a missed routine.

#445 · Control

When should a team repeat a measurement system study after an improvement?

Answer

When the process variation shrinks, the gauge becomes a larger share of the total variation and can no longer support the tighter decisions. In the chapter's example the gauge share rises from 22.5 percent to 45 percent as the process improves, which is above the usual 30 percent guideline and makes the gauge the largest single source of variation. The team repeats the study so the measurement system can be improved before the tighter limits are trusted.

#446 · Control

How does a team audit a control plan after the project team leaves?

Answer

An audit checks that the chart is being plotted on time, the reaction plan is followed, the limits still match the process, and the responsible person is actually available. A control plan that is never audited slowly becomes a wall poster instead of a working control.

#447 · Control

What does it mean to transfer ownership of a process at project close?

Answer

Transfer means the process owner accepts the new standard, the control plan, the measures, and the responsibility for holding the gain. The project team stops directing the work, and the owner gains the authority and the resources to keep the improved method running.

#448 · Control

How does a team set the sampling frequency for a new control chart?

Answer

The frequency is chosen so the process has a chance to change between samples, which means sampling quickly where the process can drift and more slowly where change takes days. Sampling too often adds inspection cost without telling the team anything new, and sampling too seldom lets a shift run for hours before the chart shows it. A bakery that changes dough mix every hour samples more often than a plant that changes its formula once a month.

#449 · Control

What is a run chart, and how does it differ from a control chart?

Answer

A run chart plots a measure over time with a center line, often the median, and it shows trends, shifts, and cycles without any calculated limits. A control chart adds control limits computed from the process variation, so it can separate common cause from special cause and signal when action is justified. A team may start with a run chart on two weeks of data and move to a control chart once it has enough subgroups to set limits.

#450 · Control

Why is the range chart read first on an X-bar and R pair?

Answer

The average chart limits are computed from the average range, so those limits hold only while the spread stays stable. If the R chart signals a change in spread, the average chart limits are no longer valid and a point there can look like a shift in the mean when the spread actually moved. Read the R chart first, fix a spread problem, then judge the average chart.

#451 · Control

What is the difference between common cause and special cause?

Answer

Common cause variation is the normal, everyday variation built into the system, and special cause variation comes from an unusual, identifiable source. A control chart signals special cause when a point falls outside a control limit or when a run or pattern rule fires. A single point far above the upper limit points to a special cause worth investigating.

#452 · Control

When does a team use an X bar and s chart instead of an X bar and R chart?

Answer

An X bar and s chart plots the subgroup mean and the subgroup standard deviation, and it becomes the better choice when subgroups are large, commonly 10 or more, because the range uses only the two extreme values of the sample. The range chart stays simple and works well for small subgroups of about 2 to 9. A process sampled in subgroups of 12 should use the s chart, since the range would throw away most of the information in each sample.

#453 · Control

What are the common runs rules for control chart analysis?

Answer

Runs rules look for patterns beyond a single point outside a limit, such as eight points in a row on one side of the center line, seven points in a row steadily rising or falling, two of three points beyond two sigma, or four of five points beyond one sigma. These rules catch small shifts and trends that a single limit check would miss. More rules raise sensitivity but also raise the chance of a false alarm.

#454 · Control

What is tampering, and why does it make a process worse?

Answer

Tampering means adjusting a stable process because of ordinary common cause variation, as if each small movement were a real problem. Each adjustment adds new variation instead of removing the cause, so the process wanders more than before. A manager who tweaks a machine every hour based on the last reading is tampering with the process.

#455 · Control

How are X bar and R chart limits calculated?

Answer

The X bar chart uses the grand average plus or minus A_2 times the average range, and the R chart uses D_4 times the average range for the upper limit and D_3 times the average range for the lower limit. For subgroups of 5, A_2 is 0.577, D_4 is 2.114, and D_3 is 0. With a grand average of 10.0 and an average range of 0.5, the X bar limits are 10.0 plus or minus 0.2885, or 9.71 and 10.29, and the R upper limit is 1.06.

#456 · Control

How do p, np, c, and u charts differ?

Answer

A p chart tracks the proportion defective and allows a changing subgroup size, while an np chart tracks the number defective with a constant subgroup size. A c chart tracks the number of defects with a constant inspection unit, and a u chart tracks defects per unit when the inspection unit can change size. A hospital tracking wrong-site events per 1,000 procedures with a changing monthly volume would use a u chart.

#457 · Control

How do autonomous maintenance and planned maintenance divide the TPM work?

Answer

Autonomous maintenance gives operators the daily tasks that keep their own equipment healthy, such as cleaning, lubricating, and noticing early signs of trouble. Planned maintenance gives the maintenance crew the scheduled work and the condition checks that need special skills, tools, or downtime. Dividing the work this way catches small problems between maintenance visits instead of letting them grow into breakdowns.

#458 · Control

Why do attribute control charts need large samples?

Answer

The limits on a p or np chart come from the binomial distribution, and the usual formulas work well only when each sample is expected to hold several defectives, often stated as an expected count of five or more. Small samples give limits that swing from subgroup to subgroup, so common cause variation looks like a signal. Auditing 200 records a day gives a readable defect proportion, while auditing 20 gives a chart that jumps on a single bad record.

#459 · Control

What is the average run length of a control chart?

Answer

The average run length is the average number of points plotted before the chart gives a signal, and for a stable process with 3-sigma limits it is about 370 points. That means a false signal appears on average about once every 370 points when only common cause variation is present.

#460 · Control

How does preventive maintenance differ from predictive maintenance?

Answer

Preventive maintenance acts on a schedule no matter what the equipment condition is, while predictive maintenance uses condition data such as vibration or temperature to decide when service is due. A rising vibration reading can trigger a bearing change before the failure stops the line.

#461 · Control

Why can an individuals chart give false signals when readings are autocorrelated?

Answer

When each reading is close to the one before it, the moving range is small, so the limits become too tight and ordinary drift looks like a special cause. A slow reactor whose next reading always follows the last one shows this pattern, and the team checks for autocorrelation before trusting the chart.

#462 · Control

When is a CUSUM chart better than a Shewhart chart?

Answer

A CUSUM chart adds the deviations from the target over time, so it detects a small sustained shift much faster than a Shewhart chart with the same false alarm rate. A process that drifts by 0.3 sigma is nearly invisible on a Shewhart chart but clear on a CUSUM chart.

#463 · Control

When does a process need a multivariate control chart?

Answer

When two or more correlated characteristics must be watched together, a chart on each one separately can miss a change in the combination. The multivariate chart plots the joint distance of the readings from the center and signals when the combination moves.

#464 · Control

How do warning limits differ from control limits on a control chart?

Answer

Warning limits sit closer to the center line, often at two standard deviations, and control limits sit at three, so the warning limits give the team an early hint. A chart can use two of three points beyond the warning limit as a rule that signals before any point reaches the control limit.

#465 · Control

Why is a chart center line not always the process target?

Answer

The center line is computed from the process data, so it shows where the process actually runs, while the target is where the team wants it to run. A chart centered at 502 grams with a target of 500 shows that the process is stable but offset.

#466 · Control

How does a controlled process differ from a capable process?

Answer

A controlled process is stable and predictable, and a capable process is also able to meet the specification limits. A process can be perfectly in control and still make out-of-specification parts, which is why both checks are needed.

#467 · Control

How do mean time between failures and mean time to repair differ?

Answer

Mean time between failures measures how long the equipment runs between breakdowns, and mean time to repair measures how long it takes to restore it. A machine with a long time between failures and a long repair time can still lose more availability than a machine that fails often but is fixed in minutes.

#468 · Control

How does a false alarm differ from a missed signal on a control chart?

Answer

A false alarm is a signal when the process has not changed, and a missed signal is no alarm when the process has shifted. Tightening the limits reduces missed signals and increases false alarms, so the limits are set to balance the two costs.

#469 · Control

How does a team chart a process that has a strong seasonal pattern?

Answer

When the seasonal swing is real and expected, the team compares each reading with the same period in prior cycles, or models the season separately, so the seasonal change is not treated as a special cause. A utility that charts December demand against November demand would alarm every year.

#470 · Control

What is an exponentially weighted moving average chart, and when does it beat a Shewhart chart?

Answer

An exponentially weighted moving average chart plots a weighted average of all past points, with recent points carrying the most weight through a smoothing constant such as 0.2. It reacts faster than a Shewhart chart to a small shift in the mean, such as a drift of one standard deviation, while the Shewhart chart is quicker at large jumps. Teams use it for slow drift, and they still plot the individual values underneath.

#471 · Control

What is pre control, and how does it differ from a control chart?

Answer

Pre control divides the tolerance into zones around the target and tells the operator to check again when a reading lands in a warning zone and to adjust after a repeated warning, using five consecutive green readings to start production. It is a setup and go or no go method based on the specification, while a control chart judges the process against limits calculated from its own variation. Pre control is simple enough for the line, but it uses the specification rather than the voice of the process.

#472 · Control

What should an out of control action plan contain?

Answer

The plan names who responds to a signal, what data they check first, which process adjustments are allowed, and when to stop the process and call for help. A chart on a fill line might say that the operator checks the last five fills, verifies the valve setting, and calls maintenance if the shift is confirmed. Writing the response in advance keeps the reaction consistent across shifts and keeps the chart from becoming a decoration.

#473 · Control

When should a team recalculate control limits?

Answer

A team recalculates limits after a permanent, documented change to the process, when the original limits rest on too few subgroups, or when a signal has been investigated and traced to a cause that the team then removed. Limits are never recalculated just to keep points inside them, because that hides the very signal the chart exists to show. The reason for the new limits goes into the control plan so the change is traceable.

#474 · Control

How does subgroup size change the sensitivity of a control chart?

Answer

A larger subgroup narrows the control limits because the standard error of the plotted average shrinks as the subgroup grows. A one standard deviation shift in the mean stands out sooner on a chart built from subgroups of 30 than on one built from subgroups of 5. Larger subgroups also cost more to collect and can hide variation that occurs inside the subgroup, so the team balances sensitivity against the sampling plan.

#475 · Control

What are standard operating procedures and work instructions, and why do they matter to a control plan?

Answer

A standard operating procedure describes how a process is run, and a work instruction gives the detail needed to perform one task the same way every time. When the improved method lives only in the team's memory, a new operator will drift back to the old method within weeks. The documents are updated during the Control phase and become the reference that the process owner trains against.

#476 · Control

How does a team replicate a proven improvement at another line or site?

Answer

Replication means applying the proven method to another line, site, or product family, and the documented standard, the control plan, and the training have to travel with it. The receiving site then confirms the gain in its own data rather than assuming the result will transfer unchanged. A fix that worked on one bottling line still needs a check on the second line, because its equipment and demand pattern differ.

#477 · Control

What is containment, and where does it sit in a reaction plan?

Answer

Containment is the immediate step that protects the customer while the cause is investigated, such as quarantining suspect product or screening the next lots. It is temporary and it does not remove the cause, so it comes before the investigation and the permanent corrective action. The reaction plan names the containment step first, because separating good product from bad cannot wait for the root cause study.

#478 · Control

What do the stratification and mixture runs signals mean?

Answer

Stratification is a long run of points hugging the center line, and it usually means the limits are wider than the current data warrant, so the chart is slow to react. A mixture is a long run of points that stay away from the center line on both sides, which usually means two different streams or processes are plotted together. Both signals point at how the data are grouped rather than at one bad point, so the team reviews the subgrouping and the limits.

#479 · Control

How does a control chart show that an improvement is holding?

Answer

After the improvement, the team compares the new chart with the baseline period and looks for a sustained shift in the average, a smaller spread, and an absence of the special causes that were common before. If the defect count fell from 12 per week to 4 per week and stays near 4 for three months, the gain is visible in the data rather than only in the project report. The chart also shows drift early enough to correct it.

#480 · Control

What is the difference between a corrective action and a preventive action?

Answer

A corrective action removes the cause of a problem that already happened, while a preventive action removes the cause of a problem that could happen. After a labeling error reaches a customer, the team corrects it by retraining and adds a preventive check that stops the wrong label from being printed at all. Both actions belong in the control plan, and each one needs an owner and a completion date.

#481 • Design for Six Sigma

What is DMADV, and how does DMADOV differ?

Answer

DMADV stands for define, measure, analyze, design, and verify, and it is the main Design for Six Sigma roadmap for a new product or process. DMADOV inserts an optimize step between design and verify, so the design is tuned before it is validated. A team designing a new infusion pump follows DMADV to set requirements, create design options, and verify performance against the customer needs.

#482 • Design for Six Sigma

When should a team use Design for Six Sigma instead of DMAIC?

Answer

DMAIC improves a process that already exists, while Design for Six Sigma creates a new product, service, or process or redesigns one so completely that little of the old design survives. Design for Six Sigma starts with customer requirements and builds quality in rather than finding defects later. A team facing a brand new product line uses Design for Six Sigma, while a team fixing late deliveries in an existing line uses DMAIC.

#483 · Design for Six Sigma

What does design for X cover?

Answer

Design for X means designing with a specific goal built in, including design for cost, design for manufacturability or producibility, design for test, and design for maintainability. Each one pushes the design team to consider a constraint early, when changes are cheap. A design that needs a special fixture to assemble is a producibility problem that is far cheaper to fix on the drawing than on the line.

#484 · Design for Six Sigma

What are robust design, tolerance design, and statistical tolerancing?

Answer

Robust design makes product performance insensitive to variation in noise factors such as temperature, operator, or raw material, so the product works well even when conditions are not ideal. Tolerance design sets the allowed variation for each part so the total variation still meets the requirement, and statistical tolerancing predicts the combined variation from the parts instead of assuming the worst case. Two parts that are each plus or minus 0.1 mm stack to 0.2 mm in the worst case but to about 0.14 mm using the root sum square method.

#485 • Design for Six Sigma

What are the deliverables of each DMADV phase?

Answer

Define produces the project charter, the business case, and the project goals, and Measure produces the customer requirements, the critical to quality characteristics, and the design requirements. Analyze develops and tests design concepts, Design produces the detailed design and the control plan, and Validate proves the design meets the requirements in a pilot or a production trial.

#486 • Design for Six Sigma

What does the optimize phase in DMADOV actually tune?

Answer

The optimize phase finds the settings that make the design perform well and stay robust across expected variation. A team that has a working prototype can tune material, dimensions, or process settings until the response is centered and insensitive to the noise factors the customer will create.

#487 · Design for Six Sigma

What is design for producibility, and what does it prevent?

Answer

Design for producibility shapes the product so the existing equipment, tolerances, and assembly methods can make it reliably at the required rate. It prevents a design that works in a prototype shop but needs a new machine, a tighter tolerance, or a hand operation on the production line.

#488 · Design for Six Sigma

What are noise factors in robust design, and how does tolerance design respond to them?

Answer

Noise factors are sources of variation the designer cannot control, such as room temperature, user habits, or incoming material differences. Robust design first makes performance insensitive to those factors, and tolerance design then sets the allowed range for each part, using statistical tolerancing to combine them rather than assuming every part sits at its worst limit.

#489 · Design for Six Sigma

What are the DMADV phases?

Answer

DMADV stands for define, measure, analyze, design, and validate. It is used to create a new product or process or to redesign one when the current process cannot reach the needed performance. The define phase sets the goal, measure captures customer needs and CTQs, analyze studies design options, design builds the chosen solution, and validate proves it works.

#490 · Design for Six Sigma

What is DMADOV, and how does DFSS differ from DMAIC?

Answer

DMADOV adds an optimize phase after design, so the roadmap is define, measure, analyze, design, optimize, and validate. Design for Six Sigma uses DMADV or DMADOV to build a new process that meets requirements from the start. DMAIC instead improves an existing process, so DMAIC is not a Design for Six Sigma method.

#491 • Design for Six Sigma

What is design for test, and what does it add to a design?

Answer

Design for test builds inspection and test needs into a product from the start, so a fault can be found quickly and cheaply instead of only after the product is assembled or shipped. It may add test points, built-in self-checks, or a diagnostic port, and it trades a small unit cost for much lower repair and warranty cost. A circuit board without test points can force a technician to unsolder parts just to find one bad connection.

#492 • Design for Six Sigma

What are the inner and outer arrays in a robust design experiment?

Answer

In robust design the inner array holds the controllable factors and their settings, while the outer array holds the noise factors that the design cannot control, such as temperature, humidity, or user behavior. Running each inner array setting against every noise condition shows which settings hold performance steady across the noise, and the team then picks the setting with the smallest variation around the target. A team testing four control settings against three noise conditions runs 4 times 3, or 12, trials.

#493 · Design for Six Sigma

What is a concept selection matrix used for in a design project?

Answer

The matrix compares several design concepts against a reference concept on a set of criteria, and the team marks each concept better, same, or worse. Adding the marks shows which concept wins on balance before any detailed design work is done.

#494 · Design for Six Sigma

What does a CTQ flow-down do in a design project?

Answer

The flow-down takes each customer critical-to-quality requirement and translates it into design parameters, then into the tolerances and process settings that control them. A requirement for a quiet pump flows down to a shaft alignment tolerance on the drawing.

#495 • Design for Six Sigma

How does redundancy change the reliability of a system?

Answer

Reliability falls fast when parts are in series, because the system works only if every part works, while a parallel or redundant path raises reliability because the system works if either path works. Two independent paths that each work 90 percent of the time give 81 percent in series and 99 percent with a parallel backup.

#496 • Design for Six Sigma

What is accelerated life testing in a design project?

Answer

It runs the product at higher stress than normal use so failures appear in weeks instead of years, and the results are then translated back to use conditions. A seal tested at high temperature can reveal a wear mode long before a field test would.

#497 · Design for Six Sigma

What does each phase of DMADV deliver?

Answer

Define delivers the design goals and the business case, Measure delivers the customer requirements and the design specifications, Analyze delivers the concept designs and the chosen concept, Design delivers the detailed design and the prototype, and Validate delivers the pilot results and the proof that the design meets the requirements. A new pump might move from a target flow rate, to a chosen impeller concept, to a tested prototype that holds the rate across the full temperature range. Each phase ends with a review before the next one starts.

#498 · Design for Six Sigma

How do verification and validation differ in a design project?

Answer

Verification checks that the design meets its written requirements, while validation checks that the finished product does what the customer actually needs in real use. A verification test might show that a device holds its temperature within one degree in the lab, and a validation test shows that it still holds that temperature on a hospital floor. A design can pass verification and still fail validation when the requirements themselves were wrong.

#499 • Design for Six Sigma

What is design for manufacturability, and how does it lower cost?

Answer

Design for manufacturability shapes the product so the existing process can build it with less effort, fewer operations, and fewer chances for error. Reducing a housing from 14 fasteners to 6 and using one shared screw size lowers assembly time and the number of parts a worker can grab by mistake. The savings come from changes made on the drawing, which cost far less than changes made after the tooling is cut.

#500 • Design for Six Sigma

What is design for maintainability, and how does it change a design?

Answer

Design for maintainability arranges the product so routine service can be done quickly, safely, and with common tools. Moving a filter to the front panel and using one fastener size might cut a scheduled service from 90 minutes to 45. The design team sets repair time and access goals early, because maintainability is expensive to add once the layout is fixed.